

European Network of
Transmission System Operators
for Electricity

ENTSO-E

CGM BUILDING PROCESS IMPLEMENTATION GUIDE

DC PART

FOR SYSTEM OPERATIONS

1.0 EDITION

18 NOVEMBER 2020

COMMON GRID MODEL BUILDING PROCESS (CGM_BP)

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REVISION HISTORY

Version	Release	Date	Comments
0	1	2020-06-16	Initial draft
0	1	2020-06-17	Section added for inclusion of DC changes in IEC 61970-600 and IEC 61970-301 Edition 7.
0	2	2020-06-18	Updated after comments by Maja.
0	3	2020-07-10	Updated after internal review
0	3	2020-09-01	Updated after researching HVDC control functionality and internal discussions at CGM BP meetings.
0	3	2020-09-02	New section on modelling options
0	3	2020-09-03	Discussion on the converter modelling added
0	4	2020-09-09	This document replaces all previous versions
0	4	2020-09-23	Updated with review comments from 2020-09-11
0	5	2020-09-25	Updates according to CGM BP meeting 2020-09-25
0	5	2020-10-19	More updates according to CGM BP meeting 2020-09-25 and subsequent discussions.
0	6	2020-10-28	New draft started from clean 0.5 2020-10-19
1	0	2020-11-18	Updated after review comments, final version.

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1. Introduction

1.1 Overview

This document, the CGM BUILDING PROCESS IMPLEMENTATION GUIDE DC PART (DC IG) complements the CGM BUILDING PROCESS IMPLEMENTATION GUIDE AC PART (AC IG) [6] with HVDC models for the CGM building process (CGM BP). The AC IG [6] is an important basis for this document and readers are presumed to be acquainted with the content of AC IG. Several restrictions and additional rules on how to model HVDC are described in this document. As this is not a formal specification the restrictions and rules from this document will be added and formally described in an extended version of QoCDCv.3.2 [3].

This Implementation Guide describes how to exchange HVDC models based on IEC TS 61970-600:2017 [1]. It describes HVDC Link functions and modelling approaches for HVDC system in grid operation. Exchange of HVDC data in the form of boundary injections is described to some extent in AC IG [6] which is extended by this document. This kind of information about HVDC is rarely described in IEC standards or specifications. Based on the experiences from interoperability tests, restrictions on data describing HVDC Links has been identified. The HVDC model from IEC 61970-600-1/2:2021 include more explicit restrictions than IEC TS 61970-600-1/2:2017 that will be added in the coming version of QoCDCv3.2 [3] that will contain the explicit formulations of the restrictions.

The following issues have been solved in this version of the DC IG:

- Shunts or filters used by current source converters may, by design, be present in the DC or AC substation. A result of the split between DC IGMs and AC IGMs, shunts or filters may end up in either a DC IGM or an AC IGM with the consequence that voltage control in power flow will behave differently. Hence shunts and filters shall preferably be in the AC IGM, if modelled. This enable voltage control utilizing the shunts also without the DC IGM.

- A DC boundary may be defined for HVDC Bipoles or for HVDC Poles. If a boundary is defined at a HVDC Bipole, pole splitting as defined in CGMA [10] will not work. To enable a simple and consistent pole splitting DC boundaries are defined for HVDC Poles only and not for HVDC Bipoles.

- Market schedule values shall be mapped to HVDC Poles to enable the poles split and loss calculation function (PSLC).

- For power flow calculations to properly respond to contingencies or changes in active power transfer the response need to be described so it can be simulated by power flow algorithm. For HVDC Links the CIM does not sufficiently describe this and Annex A list the issues. Four levels of modelling detail have been introduced to support this as further described below.

- Current source converter filters are used both for harmonics and reactive power compensation. This document describes how this is done and the need for support by the CIM/CGMES standards.

- Issues with current CIM/CGMES standards have been identified but none of them prevent from using the DC IG. The issues are listed in Annex A.

Three concepts describe HVDC transmission at power system level, HVDC Links, HVDC Bipoles and HVDC Poles. A HVDC Link consists of one or more HVDC Bipoles or HVDC Poles where an HVDC Pole carries the active power between the rectifying and inverting converter in the HVDC

Pole.

CGM Building Process working group (CGM BP WG) has defined a modelling approach for HVDC with several levels of detail:

- Simplified HVDC model which is briefly described in AC IG [6] and extended in this document. A simplified HVDC model is exchanged as part of AC IGMs without any HVDC components.
- Detailed HVDC model which is covered in this document. A detailed HVDC model need exchange of both AC and DC IGMs.

HVDC Poles can be modelled at several levels of detail depending on how much detail a TSO has available and want to provide. The simplest model is a cim:EquivalentInjection (EI) with constant active/reactive power injections without further detail. For a voltage source converter the cim:EquivalentInjection (EI) can instead be constant active power injection and constant voltage which resembles a synchronous machine. The most detailed model includes the converters in the HVDC Pole. The following levels of modelling detail for a HVDC Pole are defined:

1. Simplified HVDC described by a cim:EquivalentInjection without voltage control and without filters and shunts. Used for current source converters. This is a simplified HVDC model that cannot be combined with a DC IGM and shall only be used if the intent is to never assemble the AC IGM with a matching DC IGM. See also section 7.2.2.
2. Simplified HVDC described by a cim:EquivalentInjection with voltage control in an AC IGM. Used with voltage source converters. This is a simplified HVDC model that can be assembled with a matching DC IGM. See also section 7.2.3.
3. Simplified HVDC described by a cim:EquivalentInjection without voltage control and with shunts and filters doing voltage control. Used with current source converters. This is a simplified HVDC model that can be assembled with a matching DC IGM. See also section 7.2.4.
4. Detailed HVDC described by a model assembled from AC and DC IGMs and with shunts and filters, if any, doing voltage control. The cim:EquivalentInjections at the two boundaries are not used. For a current source converter, the shunts and filters in the AC IGM are used for harmonics filtering and voltage control (reactive power compensation). This is a detailed HVDC model and the included AC IGM(s) shall comply with levels 2 or 3 above. See also section 7.2.5.

For a description of when the to use a particular option refer to the business process description in section 5.

1.2 Reading guide

1 Introduction

Gives an overview of the document.

2 Scope

Describe what the document covers.

3 Related Documents and 4 Definitions and abbreviations

Just what the headlines says.

5 Business process

Describe the business process.

6 Interfaces

This section describes the two main interfaces

- The boundary between the AC and DC IGMs and what the data in the IGMs at each side of the boundary should look like.
- How data from market schedules are used to create SHH inputs to power flow.

7 Network data

This section describes how the data is organised and used in power flow applications.

Each subsection is further described below

7.1 *Containment*

The containment for the converters is further detailed as current DC IGMs has shown a lot of variants.

7.2 *Levels of modelling detail*

The levels of detail introduced in section 1.1 and referred to in section 5 are elaborated.

7.3 CsConverter

The function and behaviour of current source converters and how they are handled in power flow calculations is described.

7.4 VsConverter

The function and behaviour of voltage source converters and how they are handled in power flow calculations is discussed.

7.5 Attributes exchanged in CGM BP

This section describes the attributes from IEC 61970-600:2017 [1] that are exchanged in the CGM BP for use in power flow calculations. Only attributes needed for power flow are described. The sections 7.3 and 7.4 gives the rationale behind the selection of attributes.

Annex A

This section suggests changes that are needed in the CIM and CGMES standards to better support HVDC. Support for remedial actions and system protection schemes is lacking but needed to describe reaction to contingencies.

Annex B

Described the differences between IEC TS 61970-600:2017 [1] and IEC 61970-600:2021 [7].

2. Scope

This document contains information and restrictions about what data shall be included in a detailed HVDC model exported as a DC IGM that consists of CIMXML files. It is based on IEC TS 61970-600:2017 [1] with the HVDC information model, the DC package, see details in section 7.5. This document stays within the scope of IEC TS 61970-600:2017 [1] and does not introduces new classes or attributes. Restrictions related to constraints and additional descriptions are derived based on a comparison done between IEC 61970-600:2021 [7] and IEC TS 61970-600:2017 [1], provided in Annex B. The comparison will impact the QoCDCv3.2 [3] and a revised version will be created containing business process rules further limiting data.

Multi terminal HVDC networks are outside the scope of this document. Any TSO with known future developments that might be multi terminal HVDC systems should use the simplistic method as an interim solution. Point to point HVDC Links are in scope.

Formulas that describe the converter is covered in the literature and IEC specifications. To use the formulas, the control functions that compute target values must be known. Function block models in the time domain (dynamics) exists. But simplified versions that can be implemented in steady state power flow does not exist, nor does information models describing them in CIM. For this reason, control functions not having descriptions that can be implemented in power flow are out of the scope of this document. Hence contingencies and remedial actions changing the active power transfer over HVDC Poles are not covered. For contingencies, the only covered case is a complete outage of a HVDC Bipole or monopole.

This document describes guiding information and further restrictions of current standards and specifications. The restrictions in this document are for information only and a detailed description of them will be included in an extended version of QoCDC3.2 [3].

To facilitate a transition of the contents in this document to future versions of the specifications IEC TS 61970-600:2017 and IEC 61970-600:2021 the following is noted

- Annex A suggests revisions of IEC 61970-301:2020, IEC TS 61970-600:2017 and IEC 61970-600:2021 documents.
- Annex B lists the IEC 61970-600:2021 HVDC revisions with a conclusion that applications will stay with HVDC in IEC TS 61970-600:2017 and migration to HVDC in IEC 61970-600:2021 is not necessary.
- Sections 6 and 7 may be included in a coming version of IEC 61970-301, the IEC 61970-450 series profile specifications and IEC 61970-600.

3. Related Documents

The following documents are applicable:

[1] The IEC TS 61970-600-1/2:2017 is based on the Common Information Model (CIM) 16 (UML 16v29). The following documents defines the semantic model:

- IEC 61970-301:2016 Ed6: Common Information Model (CIM) Base
- IEC 61970-302:2018 Ed1: Common Information Model for dynamics specification
- IEC 61970-452:2017 Ed3: CIM Static Transmission Network Model Profiles (i.e. the network equipment model)
- IEC 61970-456:2018 Ed2: Solved Power System State profiles
- IEC 61970-453:2014 Ed2: Diagram Layout profile
- IEC 61970-552:2013 Ed1: CIM XML Model Exchange Format
- IEC 61970-501:2006 Ed1: Common Information Model Resource Description Framework (CIM RDF) schema

[2] IEC 61970-301:2020 is the standard for the canonical information model used as basis for IEC 61970-600

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300 [3] QoCDCv3.2: The validation rules are specified in the "Quality of CGMES Datasets and
301 Calculations" (QoCDC) document, which also includes the file name convention for IGM files.

302 [4] The merging process, how to solve power flows on merged models (CGMs) and the verification
303 of a valid and safe power flow solutions are specified in the "EMF requirement specification"
304 document from 18 November 2016. The document specifies requirements needed to enable a
305 highly automated exchange process and univocal interpretation of exchange d network models
306 for the CGM process using IEC TS 61970-600-1&2:2017.

307 [5] CGMA and PEVF specifications describe the schedule data used to create the inputs for solving
308 power flows at every market time unit. The documents can be found at the ENTSO-E EDI library
309 (https://www.entsoe.eu/publications/electronic-data-interchange-edi-library/).

310 [6] AC IG: CGM BUILDING PROCESS IMPLEMENTATION GUIDE AC PART 1.3

311 [7] IEC 61970-600-1/2:2021 is the IEC standard based on CGMES 3.0. This document is in FDIS stage
312 late 2020 but will be in IS stage early 2021.

313 [8] Empty reference.

314 [9] Empty reference.

315 [10] COMMON GRID MODEL ALIGNMENT (CGMA) Application Business Requirements Specification.

316 [11] IEC 60633 Edition 2.1 Terminology for high-voltage direct current (HVDC) transmission

317 [12] Pan European Verification Function (PEVF) ENTSO-E specification.

318 [13] HVDC Transmission Power Conversion Applications in Power Systems, IEEE press and Wiley &
319 Sons.

320 4. Definitions and abbreviations

321

Term	Description
Assembled model	An IGM is assembled by pulling together CIMXML files tracing by the md:Model.DependentOn references starting from the SV CIMXML file. A CGM is assembled by pulling together the CIMXML files for the IGMs to be included in the CGM.
Boundary set	A boundary set consists of the CIMXML files EQBD and TPBD.
Market time unit	This is the term used for the schedule time steps in the documents CGMM and CACM. Same as effectiveDateTime and Model.scenarioTime in QoCDCv3.2 [3].
Bus Branch (BB) model	A network model using a simplified representation of a network. Detailed breaker bays are normally not described by cim:Switches and if cim:Switches are used they shall be flagged as retained.

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Term	Description
Node Breaker (NB) model	A network model representing breaker bays with cim:Switches flagged as non-retained. Several levels of details are possible. The simplest is just describing breakers with cim:Breakers and the more detailed is also describing isolators and disconnectors with cim:Disconnectors. Some cim:Switches may be flagged as retained in case the power flow across them need to be monitored, e.g. coupler bays.
TN	TopologicalNode represents the power flow busses in a power flow model, defined as the single fictious point in a system at which the several components of the power system like generators, loads, and feeders, etc., are connected; hence, fictious power flow busses, associated with four quantities of a load flow algorithm: voltage magnitude, phase angle of the voltage, active power, reactive power.
CN	ConnectivityNode represents the electrical connection points in a network model, is always used in NB models and could be used in BB models. Note that NB models may have parts that are BB style where detailed switch modelling has not been used.
IGM AC IGM DC IGM	Individual Grid Model. A network model provided by a Modelling Authority (e.g. a TSO). An IGM is a collection of CIMXML files for a market time unit of energy delivery day and a time horizon, e.g. one day ahead. There are two types of IGMs - An IGM describing an AC power system possibly including simplified HVDC modelling, in this document labelled as an AC IGM. - DC IGM describing a detailed HVDC converter model according to the DC information model in CIM and described by its own MAS. See also description of MAS in this table. Note that DC MAS is sometimes used to mean DC IGM. This is a misuse that is not recommended.
CGM	Common Grid Model. A power system model provided by a Merging Agent (e.g. an RSC). A CGM is assembled from IGMs for a market time unit of energy delivery day and time horizon, e.g. one day ahead.

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Term	Description
CGM BP	CGM Building Process is the process of building CGMs from IGMs.
CGM BP WG	The working group that implements CGM BP.
CGMM	Common Grid Model Methodology.
CACM	Capacity Allocation and Congestion Management
HVDC Pole	A HVDC Pole has a DC cable with a converter at each end and is the smallest unit of equipment independently capable of transferring DC power, see also Figure 2.
HVDC Bipole	A HVDC Bipole consists of two HVDC Poles.
HVDC Link	A HVDC Link consists of one or more HVDC Bipoles and HVDC Poles. Large links may have several bipoles or poles also being of different technologies, e.g. CSC or VSC.
Point of common coupling PCC	Point of common coupling: This is the point at the AC side where a converter is connected, it could be a busbar or an arbitrarily complex breaker bay that connects with converter.
PPD	Pre-processing data: a set of preliminary total net positions and unbalanced DC flows provided by a TSO to the CGMA platform to compute a reference program for the creation of IGMs for time horizons where no market data is available. From CGMA BRS [10].
PSLC	Pole split and loss calculation: a process where the flows for DC links are split on the level of boundary points with loss correction and sent to CGMA or PEVF to be validated and to generate the final reference program. From CGMA BRS [10].
“reference program” “final reference program”	A not explained term used in the CGMA BRS [10]. The result of the CGMA process is referred to as the “final reference program”. Hence it is assumed to mean - The area interchanges between TSOs are aligned. - The HVDC Pole schedules are aligned with area interchanges.
RSC	Regional Security Coordinator
OPDM	OPERATIONAL PLANNING DATA MANAGEMENT is a file exchange service for remote clients. It is used to exchange CIMXML files as well as other files.

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Term	Description
QAS	Quality Assurance Service
MAS	Modeling Authority Set that defines the origin (Modeling Authority) and purpose of an IGM.
NMD	Network Modelling Database
Reference data Master data	Reference or master data is data that describe stable information that doesn't change frequently over time, e.g. - TSO and RSC names and id's - Countries and network regions - Boundaries between network regions
1D	Day ahead process.
2D	Two days ahead process.
ID	Intraday process.

5. Business process

The business process for CGM Building Process (CGM BP) is described in the AC IG [6]. For the exchange of simplified HVDC models the business process remains unchanged. For the exchange of detailed HVDC models, a new data exchange of DC IGMs is added. Each DC IGM contain one HVDC Link that is described by its own cim:ModelingAuthoritySet (MAS) and the DC IGMs are included in the assembly process to build the CGM. Different HVDC Links shall have different cim:ModelingAuthoritySets and be exchanged as different DC IGMs. Hence it is not allowed to describe multiple HVDC Links with the same MAS nor include multiple HVDC Links in a single DC IGM. Examples of HVDC Links with several HVDC Bipoles as well as HVDC Poles is; Skagerack with three current source converter HVDC Bipoles and one voltage source converter HVDC Pole and IFA with a two HVDC Bipoles from 1986 and coming IFA2 HVDC Bipole. The HVDC Links shall be configured so they matches the scheduling.

Table 1 below show the steps to create CIMXML files for AC and DC IGMs. A table like this one and descriptions of IGM and CGM creation can be found in the AC IG [6] section 6.

Table 1 Processing steps in tabular form

#	Step description
1	The Modelling Authority (TSO) receives market schedules on daily basis (process description is outside the scope of this document). Three types of schedules are used - Schedules for interchange between scheduling areas, include both AC and DC flows. - Schedules for power transfer on HVDC Links between scheduling areas. The format of this schedule is described in section 6.4.2. - TSO internal schedules for power transfer on internal HVDC Links.

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2	Modelling Authority (TSO) creates a power system model of its area of responsibility also including neighbouring areas that impacts the internal power flow. Power flow is solved for each market time unit.
3	For the AC MAS the modelling Authority (TSO) creates one set of CIMXML files for each market time unit. Each HVDC Link has its own DC MAS. For each DC MAS the modelling Authority (TSO) creates one set of CIMXML files at a specific market time unit. A TSO that do not have a detailed HVDC model will not export CIMXML files for the DC MAS. For details on how the CIMXML files are organised refer to the corresponding table in the AC IG [6].
4	A Modelling Authority (TSO) submits the CIMXML files for validation by a Quality Assurance Service Provider.
5	A Modelling Authority (TSO) CIMXML error-free in validation levels 1-6 files are made available to Merging Agent (RSCs). If errors are detected the IGM is rejected.
6	A Merging Agent (RSC) obtains the IGM CIMXML files for Modelling Authorities (TSOs) within a region of concern. If the Merging Agent (RSC) do not support detailed HVDC models the DC IGM CIMXML files are not obtained for all DC MAS of concern.
7	The Merging Agent (RSC) creates the CGM and power flow is solved for each time unit. If the Merging Agent (RSC) support detailed HVDC models the DC IGM CIMXML files are included in the CGM.
8	The Merging Agent (RSC) provides updated power flow inputs (SSH) per Modelling Authority (TSO) and result (SV) for the CGM region for validation by the Quality Assurance Service Provider. If error-free, CGM is made available to Information receiver. If errors during validation are detected, the CGM is rejected.

The business process for HVDC data exchanges allow for different levels of modelling detail both for the exporting party (the modelling authority, a TSO), step 3 in Table 1, and the importing party (the merging agent, an RSC or TSO), step 6 in Table 1. The exporting party decide on the level of detail for the exported data as discussed in section 1 and 7.2.

A complete HVDC Pole is described by the AC parts at each side (shunts, filters, and switch gear) and optionally the DC part with converters, transformers, and cables. The AC part is exchanged in AC IGM CIMXML files and the DC part is exchanged in DC IGM CIMXML files.

An exporting party may include AC and DC parts in the export as follows

1. In case a HVDC Pole is a boundary with another TSO the export may contain
    a) One side of the HVDC Pole AC part.
    b) One side of the HVDC Pole AC part for together with the DC part.
2. In case a HVDC Pole is internal to the TSO the export options are
    a) include both sides of the HVDC Pole AC part.

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b) include both sides of the HVDC Pole AC part together with the DC part, a complete model of the HVDC Pole.

The exporting TSO have the two choices a) or b). The export option a) can use the detail levels 1, 2 or 3 from sections 1 and 7.2. The export option b) can only use detail level 4 which means that the DC part is exported together with AC part using either detail level 2 or 3.

Depending on the business process needs, the exporting party determine the level of detail for the export of the AC IGM (levels 1-3) and inclusion of the HVDC Poles in a DC IGM (level 4):

- Level 1 is suitable when active power transfer does not change and the HVDC Pole do not participate in voltage control or the impact is negligible.
- Level 2 or 3 is suitable when active power transfer does not change but response to voltage changes is wanted and the HVDC Pole or its filters/shunts need to participate in voltage control.
- Level 4 is suitable when responses to contingencies or remedial actions is wanted. This means active power transfer may change and filters/shunts need to participate in voltage control. At a contingency, the losses in the HVDC Pole may change and need to be computed.

An exporting party may not be capable to support all levels of detail which then limits the export. Likewise, an importing party may not be capable of importing data for all detail levels. Table 2 summarize the match between exporting and importing party capability to support the levels of detail.

Table 2 Summary of detail level export/import capability

	Export capability		
Import capability	AC part with level 1 detail	AC part with level 2 or 3 detail	AC and DC parts with level 4 detail
Level 1 capability	Yes	No	No
Level 2 and 3 capability	Yes	Yes	No
Level 4 capability	Yes	Yes	Yes

The top row in Table 2 describe the level of detail for HVDC Poles included in CIMXL files. The left column shows the capability of an importing party to use the HVDC Pole data. The matrix in the middle show the capability to use the data. Table 2 also show the maturity to exchange HVDC Pole data, i.e.

- If exporting parties can only provide level 1 detail this is what is needed for all parties.
- If exporting parties support level 2 or 3, the importing party need to support the same level as well as level 1.
- If exporting parties support level 4 the importing party need to support all levels 1 to 4.

For detail level 4 there are both HVDC Pole AC parts and DC parts available and a choice whether to impot the DC part can be made. Depending on the business process needs and the capability from Table 2 the options are

- When the active power transfer stays constant (e.g. contingencies and remedial actions are not considered) the losses do not change, and it is sufficient to use the HVDC Pole AC parts at detail level 2 or 3. Hence the DC part need not be imported.
- For an importing party that do not support level 4 it is still possible to import the HVDC Pole AC part at detail level 2 or 3. As a consequence changes in active power transfer cannot be managed and corresponding losses cannot be computed.
- When contingencies and remedial actions are considered, the detail level 4 DC part shall be imported and used so that the consequences of changed active power transfer can be computed as well as the corresponding losses.

How this is implemented in network models is further discussed in section 7.2.

6. Interfaces

6.1 Overview

Section 6 describes how IGMs are organized to support CGM assembly. The boundary between the IGMs and the mapping of market schedules to IGMs are also described.

6.2 Organization of AC and DC IGMs

6.2.1 HVDC Links connecting TSOs

A detailed model of a HVDC Link is in a separate DC IGM defined by a separate region under a cim:ModelingAuthoritySet (MAS). A DC IGM can only contain one HVDC Link but will contain as many HVDC Bipoles or HVDC Poles as the HVDC Link contains. A DC IGM fits with the boundary data set as any other IGM so a DC IGM can be included in a CGM assembly without adjustments of the AC IGMs it connects to.

At the DC boundary between TSOs, the shunts and filters are in the AC IGM, the converter transformer, phase reactor (if any) and the converters are in the DC IGM. A consequence is that the shunts and filters will be described in two different AC IGMs for a boundary HVDC Link, one for each TSO connecting to the HVDC Link (for internal links the same TSO is at both sides). One of the TSO will take the role of the ModelingAuthority responsible for the HVDC Link and the other will relinquish its ModelingAuthority responsibility to the responsible TSO. Both TSOs has a bilateral interest in the HVDC Link and are supposed to cooperate in creating the DC IGM despite one of them is assigned the modelling responsibility. To support this, a description of the bilateral or shared interest is needed in the information model., for more details refer to Annex A.

6.2.2 TSO internal HVDC Links

The TSO that designed and commissioned the HVDC Link may want to have shunts and filters at both sides in its own AC IGM so the HVDC Link become internal to the TSO. This is solved by considering the HVDC Link as internal and create an AC boundary in the substation managed by the other TSO. This is done by splitting the substation in two, one fictitious with the filters and shunts and the remaining substation with the rest. An AC boundary is inserted between them, see also example in section 6.3.6. Consequences of this are

- The boundary between the TSOs is an AC boundary that may be at a transmission line (exemplified in Figure 14, the common boundary).

- The TSO with the internal HVDC Link will use one of the modelling options
  - 7.2.3 for voltage source converter with AC IGM only.
  - 7.2.4 for current source converter with AC IGM only.
  - 7.2.5 for DC IGM assembled with AC IGMs.
- The HVDC schedule stays at the DC boundary and the HVDC Link is now considered internal to the TSO and all losses is taken by the TSO with the internal HVDC Link. The scheduled injection at the HVDC Link will force the same amount of active power across the AC boundary reduced with losses appearing between the AC and DC boundaries.

Detailed models for HVDC Links embedded in an AC IGM may be provided by the responsible TSO as a service to other TSOs or RSCs. Using a detailed HVDC models means including DC IGMs in the assembly of a CGM. Note that the modelling option 7.2.2 shall not be used with a DC IGM.

Scheduling of the HVDC Link power transfer is further discussed in section 6.4.3.

6.3 Boundary model between AC and DC IGMs

6.3.1 Boundary points

There are two ways to describe the HVDC boundary

- Option 1: cim:Line with two boundary points, one for each side of a HVDC Pole.
- Option 2: cim:Lines with one boundary point where each cim:Line describe one side of the HVDC Pole. The two cim:Lines has the same EIC code describing the HVDC Pole.

Option 2 makes the network model dependent on market data as the EIC code is required to find the two cim:Lines describing the HVDC Pole. The best solution is to add a new class HVDC Pole as suggested in Annex A and the second best is to use Option 1. Throughout this document Option 1 is used if nothing else is specified.

The HVDC boundary option 1 consists of one cim:Line with a pair of a cim:ConnectivityNode (CN) and a cim:TopologicalNode (TN) at each side as shown at the top in Figure 1. This is different from an AC boundary that has a cim:Line with a single CN/TN pair. Modelling a DC boundary with two cim:Lines having a single CN/TN pair is not allowed.

Boundary option 2 is shown at the bottom in Figure 1.

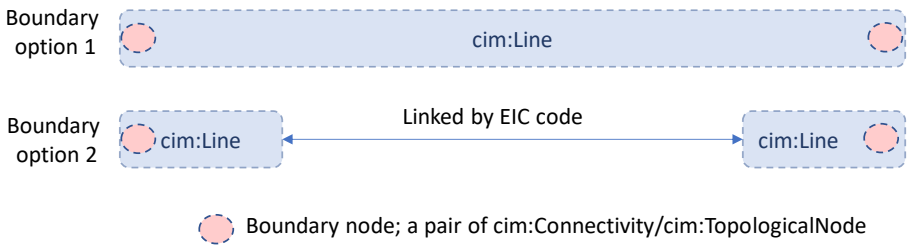


Figure 1 HVDC Boundary

Figure 1 option 1 show the same boundary as is described in the AC IG [6]. Note that for a HVDC Bipole one cim:Line shall be used for each HVDC Pole. The following boundary

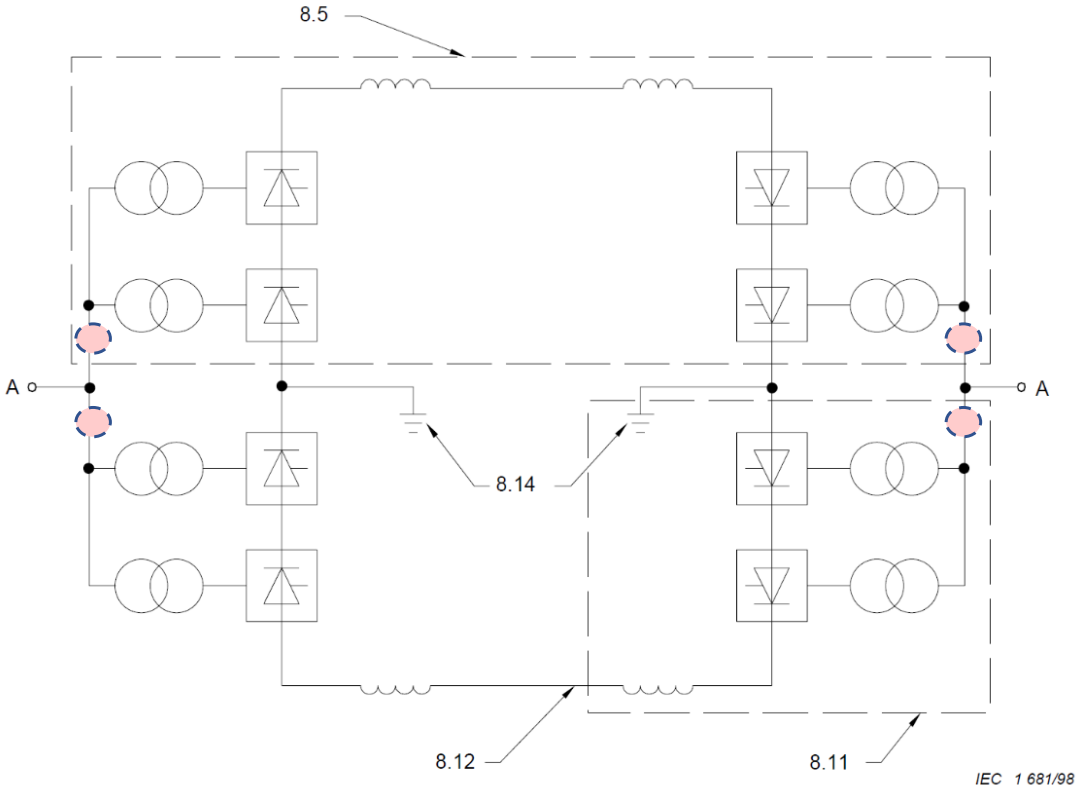
configurations are not allowed for a HVDC Bipole

- A single cim:Line with the boundary nodes linked to the two HVDC Bipoles.
- A single cim:Line with a pair of boundary nodes for each HVDC Pole resulting in four boundary nodes in total for the cim:Line.

The cim:Line is the only way in IEC TS 61970-600:2017 [1] to describe an HVDC Pole so it acts as a proxy for the HVDC Pole. Note that IEC 61970-600 allows for a boundary in a cim:Substation but this option cannot be used until the concept of a HVDC Pole is introduced and described in CIM, i.e. a class that include a description of the two sides of an HVDC Pole as the cim:Line currently does.

6.3.2 Structure of current source converter

The concept of a HVDC Pole is defined in IEC 60633 [11] and is shown in Figure 2.



Key

A	AC system	8.12	HVDC transmission line
8.5	HVDC system pole	8.14	Earth electrodes
8.11	Substation pole		

Figure 2 Definitions as defined in IEC 60633 [11]

In this document item 8.5 “HVDC system pole” in Figure 2 is called “HVDC Pole", not “HVDC system pole”. Each HVDC Pole shall have a boundary node at each side which means four boundary nodes as shown by the red circles in Figure 2.

An IGM connects to the boundary point with one branch and one cim:EquivalentInjection (EI) as defined in AC IG [6]. A branch could be

- cim:Switch contained by a cim:VoltageLevel in the AC IGM or a cim:DCConverterUnit in the DC IGM.

- cim:ACLineSegment contained by a cim:Line in the boundary.
- cim:PowerTransformer contained by a cim:Substation in the AC IGM or a cim:DCConverterUnit in the DC IGM.

The branches connecting with the boundary are in the AC or DC IGM. In the case an existing cim:Switch or cim:PowerTransformer cannot be used to connect with the boundary an artificial cim:ACLineSegment shall be used.

Three cases exist for connecting an IGM to the boundary

- Dangling boundary point where no IGM is connected appearing when IGMs at both sides of a boundary has been excluded from a CGM.
- Boundary point with one IGM connected. The EI acts as an equivalent for the not connected IGM.
- Boundary point with two IGMs connected. The two EIs are not considered.

Figure 3 shows the boundary point connection cases.

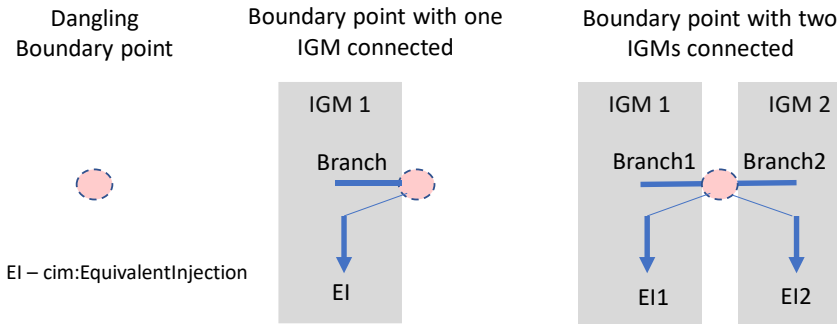


Figure 3 Boundary point connection cases

HVDC Links with more than one HVDC Pole need pole splitting, see section 6.4 for details. For a HVDC Bipole, see figure Figure 2, the active power flow divide on the HVDC Poles within the HVDC Bipole. To get a uniform pole splitting only HVDC Poles have a boundary, HVDC Bipoles and HVDC Links do not have a DC boundary.

In the case of simplified HVDC the boundary point always has a single AC IGM connected to the boundary and the EI in the AC IGM represents the HVDC converter as exemplified in Figure 4. In the case of detailed HVDC at least one side of a HVDC Pole has two IGMs connected to the boundary, one DC IGM and one AC IGM. The EI injection values shall match and be checked as further described in examples below. If the checks result in no errors the EI injection values on each side of a boundary point, shown to the right in Figure 3, are first set to zero and then updated according to the load flow result in an assembled model according to current practice in CGM BP.

A DC IGM with one side connected to its AC IGM (referred to as AC IGM1) may have the other side disconnected (referred to as AC IGM2). The EI in the DC IGM at the disconnected side represents the power exchanged with the not connected AC IGM2, it is an equivalent for it, see also example in Figure 6.

Figure 2 also shows the point of common coupling (PCC) at “A”. This connection point is not particularly useful as many branches may connect to it making it impossible to distinguish the

power flows in the connecting branches. To avoid this in Figure 2 the two HVDC Poles are given their own boundary nodes which have the PCC at each side of the HVDC Pole. The boundary points are indicated by the red circles in Figure 2. This is also the reason why the cim:ACDCConverter.PccTerminal is referring to a cim:Terminal and not the PCC node that is a cim:ConnectivityNode/cim:TopologicalNode. Voltage and current measurements used to control a HVDC Pole are located at a branch carrying the AC current flowing between the AC substation and the converter. In Figure 2 this is the line between the two transformer legs and the point "A" where the two boundary nodes are located. The measurements are typically located at the transformer side of the line.

A HVDC Link is more complex than showed in Figure 2 and the figures in the following subsections show examples with more details. As shown in Figure 3 branches and cim:EquivalentInjections (EI) are connected to the boundary point, a cim:ConnectivityNode and a cim:TopologicalNode. The cim:ACDCConverter.PccTerminal shall refer to the cim:Terminal at the branch connecting to boundary point. This make it easy to find cim:ACDCConverter and update its active power attributes at the same time as the cim:EquivalentInjection active power is updated from the market schedule.

6.3.3 Example CsConverter monopole case with shunts in AC Substation

This section gives an example of a monopolar HVDC Link including HVDC with a CGM including the AC IGMs at each side of the DC IGM.

An example of two AC IGMs with simplified HVDC in a CGM is shown in Figure 4.

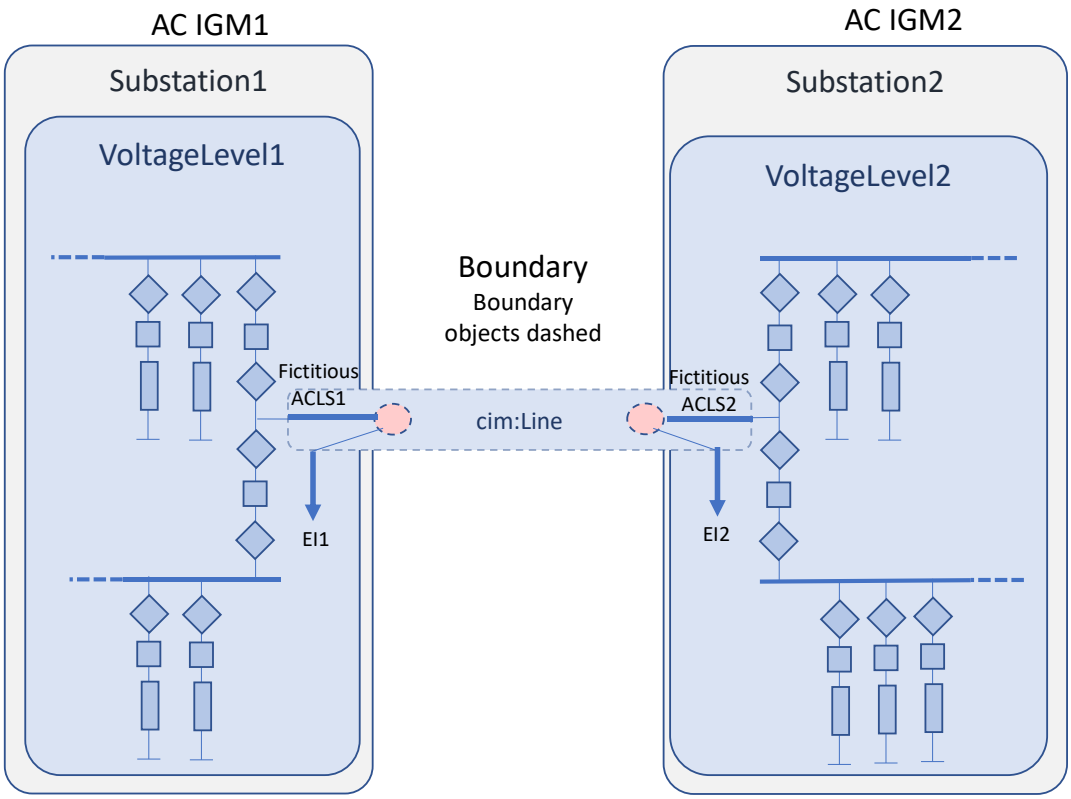


Figure 4 Example of simplified HVDC boundary

Figure 4 shows two AC IGMs connecting to the boundary nodes at each side of the cim:Line. IEC 61970-301:2020 [2] figure 38 is similar to Figure 4 and provide the same information. In

the example each AC IGM has a fictious cim:ACLineSegment (ACLS1 and ACLS2) connecting to the boundary point and one cim:EquivalentInjection (EI1 and EI2) representing the power flow into the HVDC Pole. ACLS1 and ACLS2 are fictious and introduced to have the required branch connecting to the boundary point, if a real breaker or disconnector connects to the boundary it should be used instead of the fictitious cim:ACLineSegment. If there is a cable between the AC substation and the converter the cim:ACLineSegment is not fictitious but real. A fictitious cim:ACLineSegment shall have a small series reactance greater than zero compliant with the QoCDCv3.2 rule ACLineSegmentX.

The active power flow at the two sides shall be with different signs and differ no more than the losses in the HVDC Pole. For validation purposes the losses can be assumed to be less than x% of the power flow where the x value will be determined in a revised version of QoCDCv3.2 [3]. A voltage source converter may not transfer any active power. In this case both sides will consume active power due to the losses which means they both acts as rectifiers.

An example of a detailed DC IGM is shown in Figure 5.

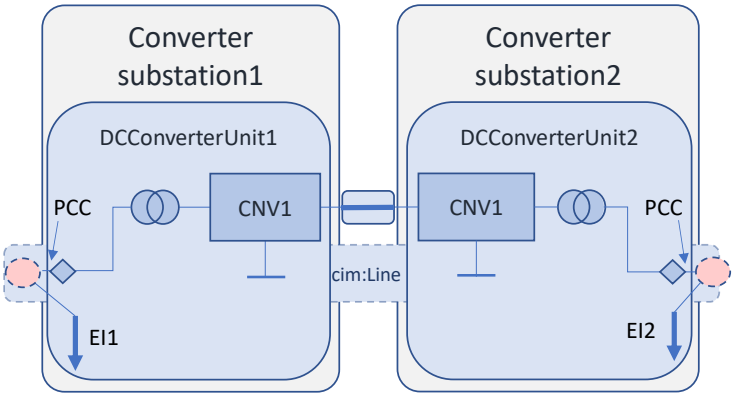


Figure 5 Example of a detailed monopole DC IGM

Figure 5 shows how the converters (cim:CsConverter or cim:VsConverter), at the two sides of the HVDC Pole are connected via a transformer to the boundary point, the point of common coupling (PCC), that connects with the AC system. In this case the cim:ACDCConverter.PccTerminal shall refer to the cim:Terminal at the boundary node side of the cim:Switch. IEC 61970-301:2020 [2] does not have a similar figure. The cim:Line with the boundary nodes in the dashed box in the background is the boundary cim:Line showed in Figure 1. The HVDC Link in Figure 5 show a monopole with one HVDC Pole and ground return.

Each side of the HVDC Pole is connected to the PCC boundary node via a cim:Switch (a breaker or disconnector) that is the branch connecting to the boundary point. As a branch is anyway needed it is advised to include any existing cim:Switch in the model, otherwise a fictitious branch, e.g. a cim:ACLineSegment is needed. The cim:EquivalentInjections EI1 and EI2 represent the power flow into the AC systems. The EI injection values shall have opposite signs and differ no more than the losses in the HVDC Pole. A voltage source converter may not transfer any active power and if so both sides of the HVDC Pole will consume active power due to the losses. A back to back system has the same interface with the AC system as shown in Figure 5 with the exception that the two HVDC cables between the converters in Figure 5 are excluded and the converters are directly connected, an example figure for this case is not included.

An example case with a detailed DC IGM assembled with one AC IGM is shown in Figure 6. The use case in Figure 6 is useful when the detailed voltage control of the remote end (at the right

side) is of no interest. It is then sufficient to assume that the active and reactive power exchange with the remote AC IGM is described by the cim:EquivalentInjection, EI2 in Figure 6, and the remote AC IGM is not needed.

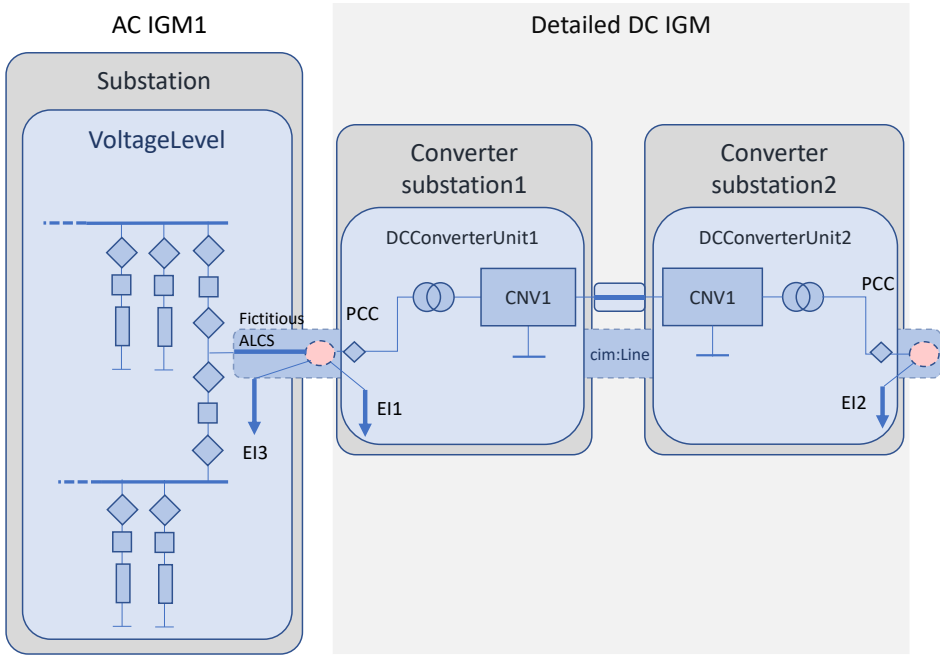


Figure 6 Example with assembly of one AC and one DC IGM

Figure 6 shows how one AC IGM1 connects with a DC IGM. IEC 61970-301:2020 [2] does not have a similar figure. The left side of the boundary has two cim:EquivalentInjections EI1 and EI3 that shall be equal with opposite signs (EI1.p + EI3.p = 0). If they are different this means an imbalance between the solutions for the AC IGM and the DC IGM which needs correction. The values are originally from market or TSO schedules and if they differ the schedules may not have been aligned as described in the CGMA document [10]. The right side has a cim:EquivalentInjection EI2 that represents the power flow into the absent AC IGM2. The converter control variables in the SSH CIMXML file shall agree with the injection values at EI3/EI1 and with EI2 including the losses.

The last example is the case where a DC IGM is connected at both sides as shown in Figure 7. IEC 61970-301:2020 [2] does not have a similar figure.

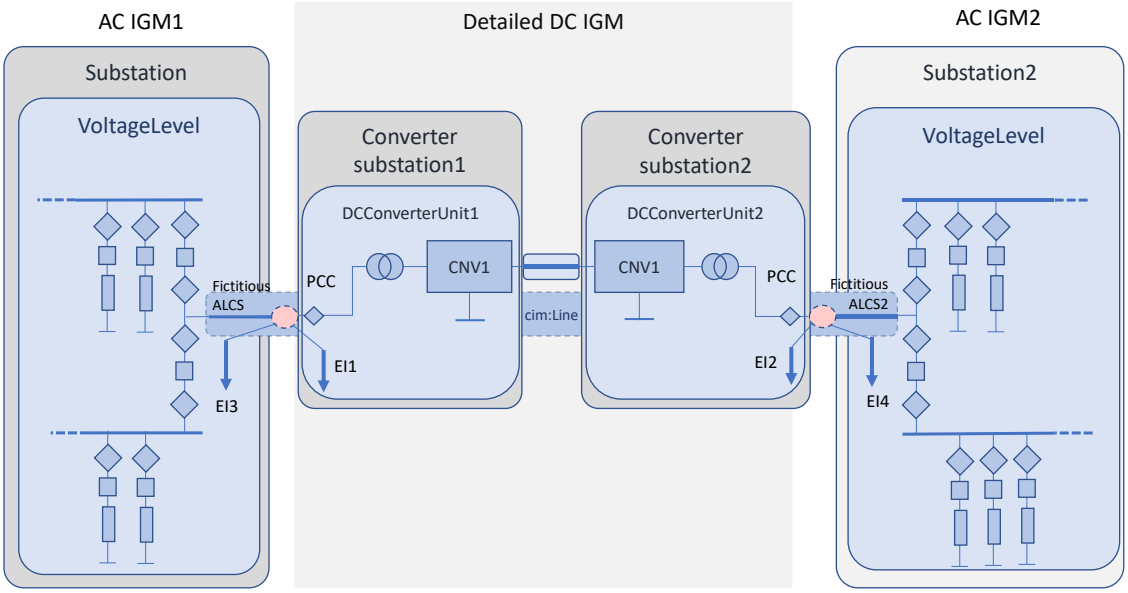


Figure 7 DC IGM connected at both sides

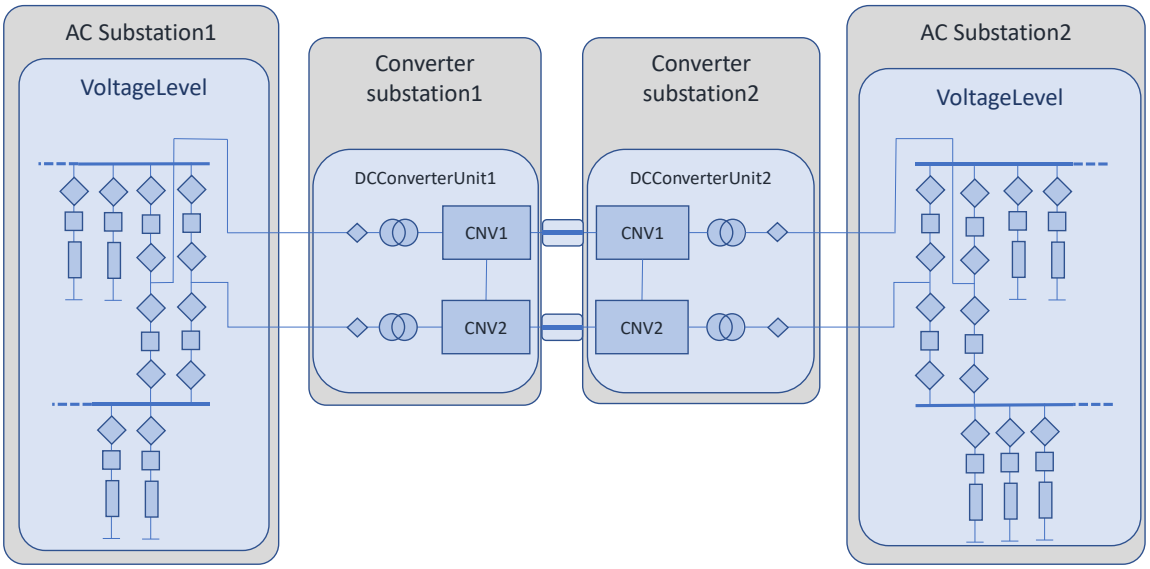
In Figure 7 the power flow through the DC link is determined by the control variable specified in the SSH CIMXML file for the converters as well as the initial EI injection values from the IGMs. If they are different this means an imbalance between the solutions for the AC IGMs and the DC IGM which needs correction. After solving power flow for the CGM the injections EI1, EI2, EI3, EI4 and converter control variables are updated with the result and shall agree with each other.

Note that converter substations in Figure 6 and Figure 7 do not have a cim:VoltageLevel and that the shunts are located in a cim:VoltageLevel within the AC substation according to modern converter substation designs. This means that AC equipment within a cim:DCConverterUnit (e.g. the cim:Disconnector and the two cim:PowerTransformers in the figures) need a reference to a cim:BaseVoltage, which is preferably defined in the equipment boundary file. Older converter substation designs may have the shunts in the converter substation, this case is further discussed below.

6.3.4 Example CsConverter bipole case with shunts in AC Substation

This section gives an example of a HVDC Bipole showing the CGM in Figure 8.

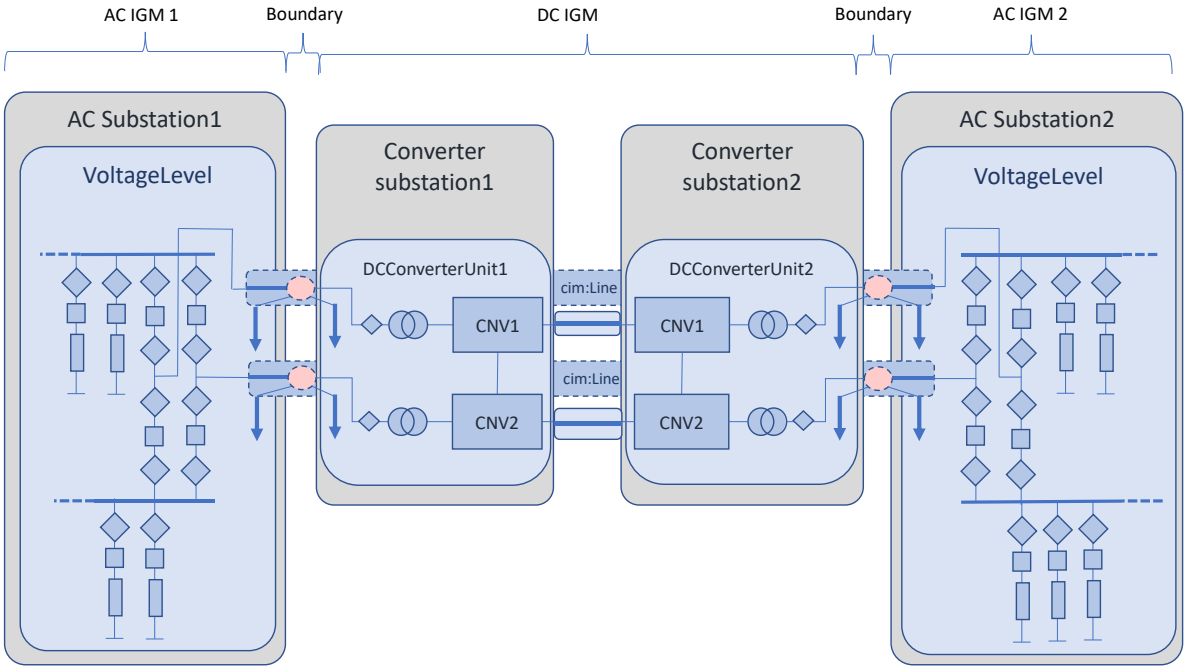
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602

603 **Figure 8 Example Bipole HVDC Link with shunts in AC substation**

604 Figure 8 shows how the two HVDC Poles in the HVDC Link are connected in the AC Substations,
605 a common way to build modern HVDC Links. Forcing a single boundary node for the bipole is
606 not possible due to the substation design (a HVDC Bipole shall not have a boundary anyway).
607 Instead two boundary nodes per side and HVDC Pole is needed. The example in Figure 8 with
608 boundary nodes and boundary objects is shown in Figure 9.



609

610 **Figure 9 Bipole from example in Figure 8 with added boundary**

611 Figure 9 show how the two HVDC Poles are connected to the AC Substations with their
612 boundary nodes. The PCC shall refer to the cim:Terminal at the boundary node side of the
613 cim:Switch for each HVDC Pole. IEC 61970-301:2020 [2] has a figure 38 showing the case where

the HVDC Bipole has a common boundary for the two HVDC Poles, this is not correct. The two boundary nodes at each of the HVDC Pole has its own cim:Line which means there is a one to one correspondence with a HVDC Pole and a cim:Line. It is impossible to have a common boundary point for the two HVDC Poles as they are separately connected via separate bays to the bus bars in the AC Substation.

6.3.5 Example CsConverter bipole with filters in DC Substation

An example of a HVDC Link with two bipoles connected to the AC substation with two separate cables is shown in Figure 10, only one half of the HVDC Poles are shown.

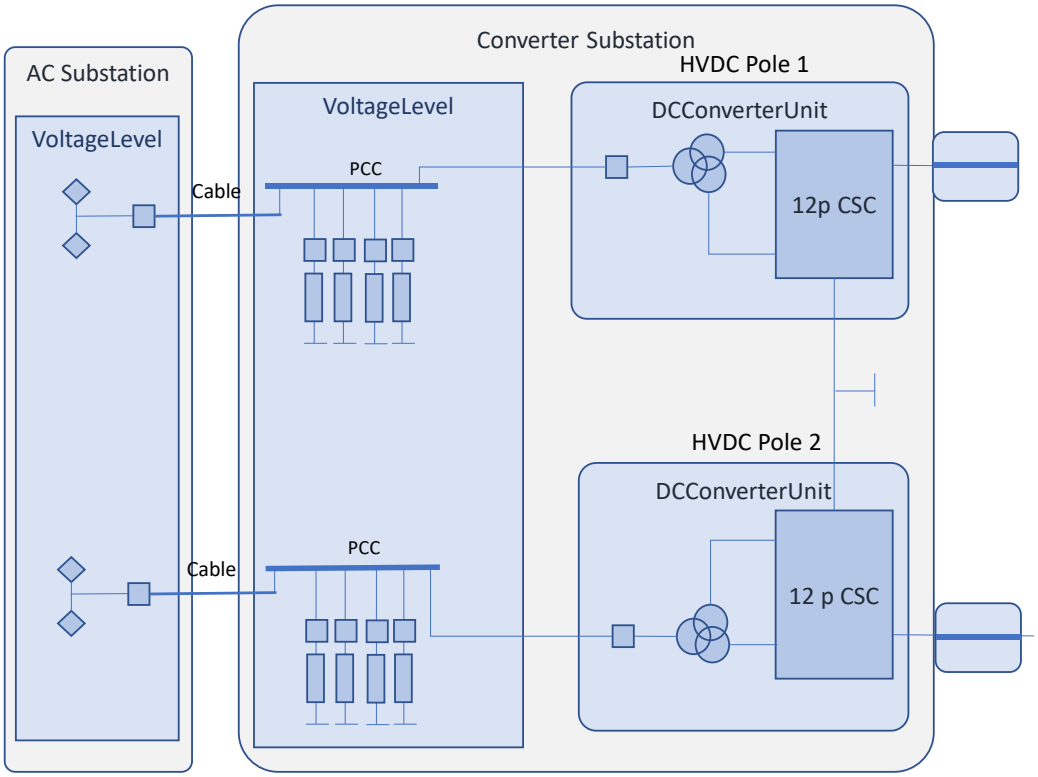


Figure 10 Example Bipole with shunts in DC substation

Figure 10 shows the HVDC Link with shunts and filters in the converter substation which may be the case for older HVDC Links. The HVDC Poles are 12 pulse converters which means they have two AC connections, see also section 7.3.1. There are two possible boundary locations in Figure 10

- Between the cim:VoltageLevel in the converter substation and the cim:DCConverterUnit. In this case the boundary is located at the converters in the HVDC Poles.
- Between the AC substation and the converter substation. If the HVDC Poles in a bipole has a common connection with the AC Substation this location cannot be used as a boundary but this is not the case in Figure 10 so this location is possible. But the location is less desirable as voltage control behavior is inaccurately modelled.

In the first case the converter substation must be split into one fictitious substation with the filters/shunts and the converter substation with the remaining cim:DCConverterUnits as shown in Figure 11.

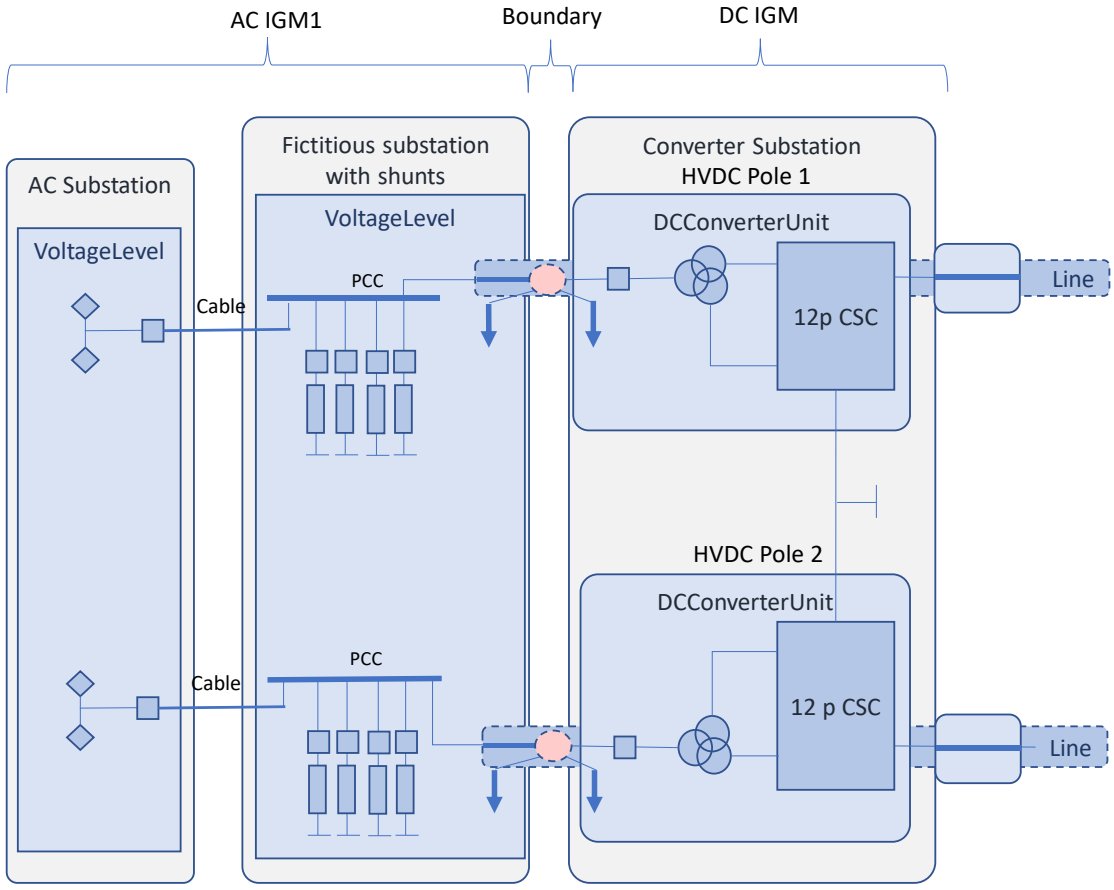


Figure 11 Bipole from example in Figure 10 with boundary between shunts and converter

Figure 11 shows how the two HVDC Poles in each bipole are connected to the fictitious AC Substation with their boundary nodes between the filters and the convertes. The PCC shall refer to the cim:Terminal at the boundary side of the cim:Switch. This enables pole splitting per HVDC Pole. The shunts/filters will be at the AC side and connected via the boundary with the converter which enables voltage control using the filters without the DC IGM.

The fictitious substation is needed because the boundary model in IEC TS 61970-600:2017 [1] allow a boundary only within a cim:Line. Note that the fictitious substation needs its own mRID, the mRID from the original substation cannot be reused.

For each pole of a bipole the boundary contain two cim:Lines (or HVDC Poles) with a pair of CN/TN boundary nodes each, according to the first option in section 6.3.1.

The case with the boundary between the AC and DC substations is shown in Figure 12.

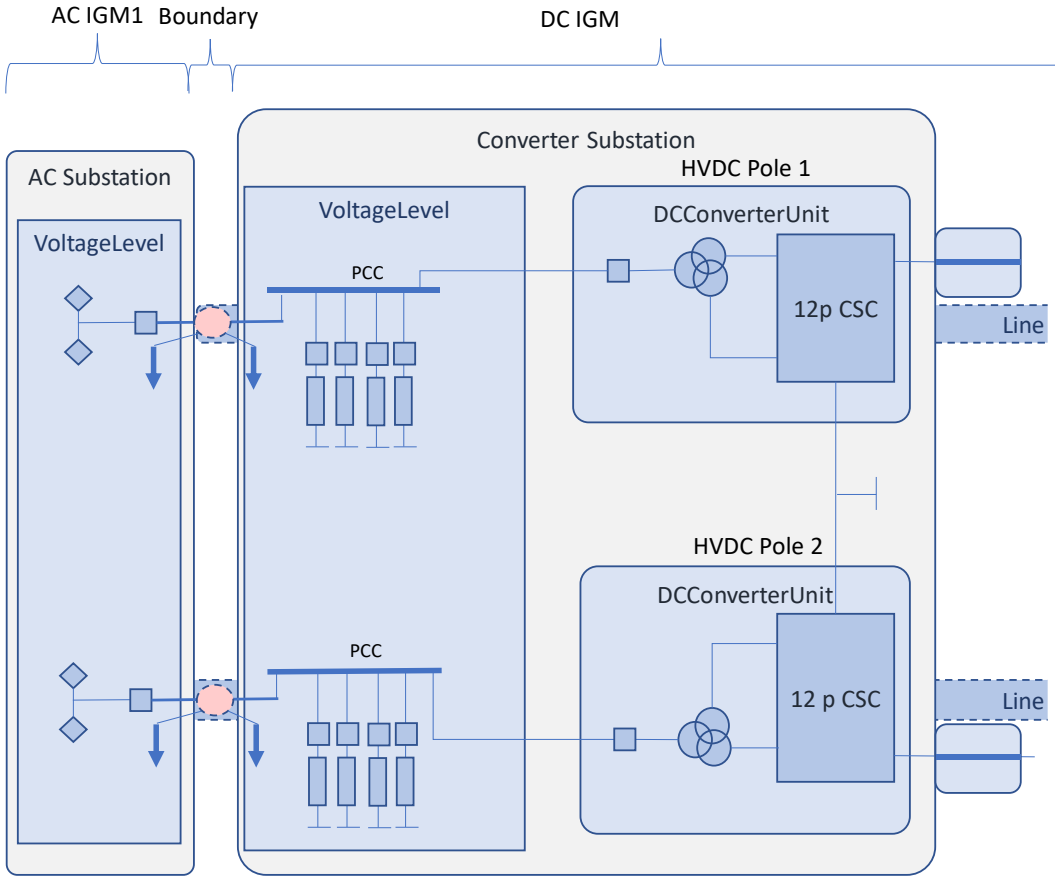
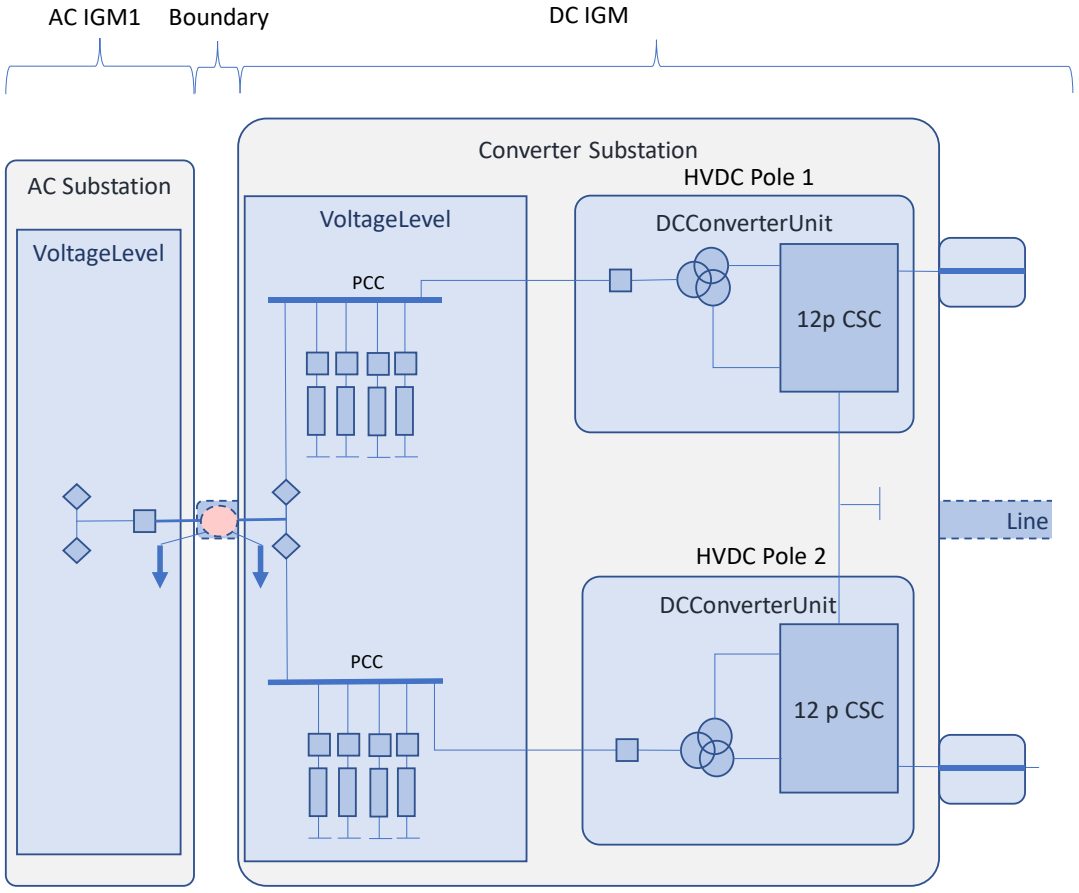


Figure 12 Not recommended HVDC Bipole boundary from example in Figure 10

Figure 12 shows the boundary where the cables are connected to the AC Substation which avoids creating the fictious substation in Figure 11. Having a boundary configuration like this is not recommended as the voltage control behave differently with and without the DC IGM. If the DC IGM control the voltage in the AC substation the cim:EquivalentInjection in the AC Substation may have voltage control enabled to simulate the DC IGM if not assembled. But this is a continuous control and not discrete as with the DC IGM. Another issue is to determine what filters are available for voltage control which impacts the cim:EquivalentInjection reactive power limits.

If the two boundary nodes in Figure 12 are replaced with a single boundary node at points “A” in Figure 2, the active power flow in the bipole is the sum of the flow in the two individual HVDC Poles which means that the cim:Line represents the bipole HVDC Link rather than each HVDC Pole. IEC 61970-301:2020 [2] figure 38 has a similar figure providing the same information. Figure 13 show Figure 12 where the boundary points has been replaced by a single boundary point at each side similar to point “A” in Figure 2.



667

668 **Figure 13 Not allowed HVDC Bipole boundary from example in Figure 10**

669 Having a boundary configuration as in Figure 13 is not allowed as discussed in section 6.4.

670 ### 6.3.6 Example CsConverter bipole with AC interface

671 This section shows examples where a HVDC Link is made internal to an AC IGM and the DC
672 boundary is internal to the AC IGM as discussed in section 6.2. For an internal HVDC Link the
673 active power transfer cannot be related to any specific nodes in the AC boundary without
674 solving power flow. Power flow solve the total exchange between cim:ControlAreas and an
675 internal HVDC Links will impact at which AC boundary nodes and tie lines the power is
676 exchanged.

677 Figure 14 shows the special case where the AC boundary is at two cim:ACLineSegments
678 connecting to the AC part of a HVDC Link.

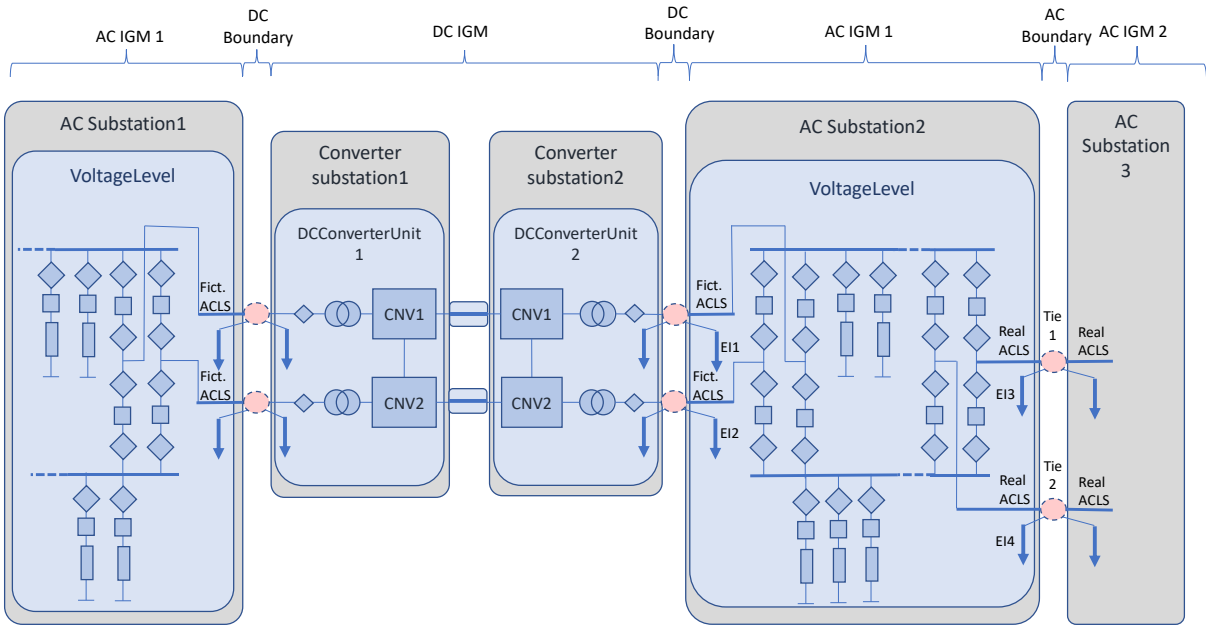


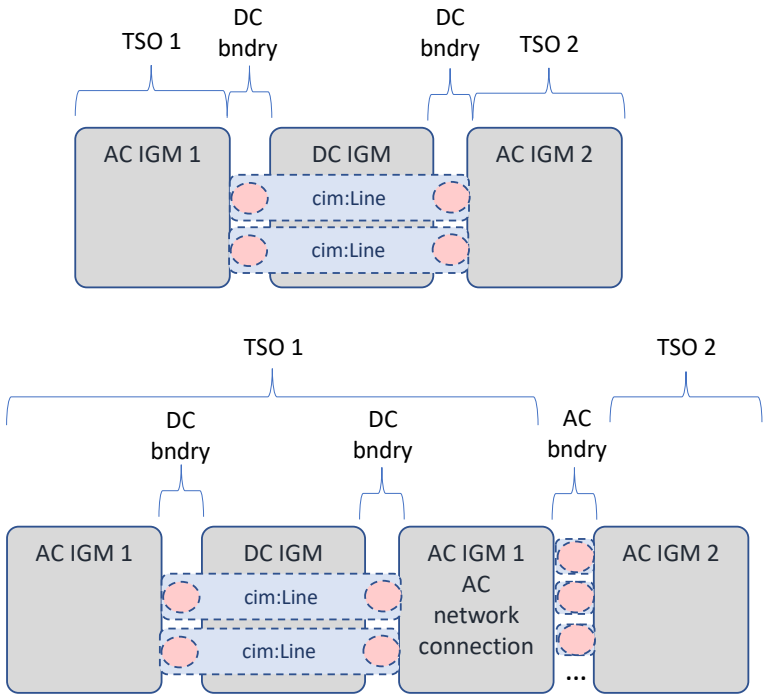
Figure 14 Example bipole with AC boundary at AC Line

Figure 14 shows a detailed DC IGM embedded in an AC IGM 1. The network between AC IGM 1 and AC IGM 2 is two cim:ACLineSegments shown to the right in Figure 14. The thick lines labeled "Fict. ACLS" are fictitious cim:ACLineSegments and the ones labeled "Real ACLS" are the real cim:ACLineSegments. The schedules for the HVDC Poles could be mapped to the AC boundary points Tie1 and Tie2 in Figure 14 but it is not always possible to easily compute the exchange at the tie points Tie1 and Tie2 . The solution is to assign the schedule values to the internal HVDC boundary (at the converter substations) and have the power flow solving the exchange over the AC boundary while keeping the net interchanges between the two cim:ControlAreas.

6.4 Market schedule interface

6.4.1 Overview

A HVDC Link between two TSOs is modelled as a DC IGM connecting two different AC IGMs, the DC IGM is then part of the boundary. A HVDC Link inside a TSO network, is modelled as a DC IGM with both ends of the HVDC Link connecting to the same AC IGM describing the TSO network (internal HVDC Link). The boundary between the TSOs is in then an AC boundary. Figure 15 exemplify the cases for a HVDC Bipole, the two HVDC Poles are described by one cim:Line (Option 1 in section 6.3.1) each which is needed for mapping market schedules to CGMES data.

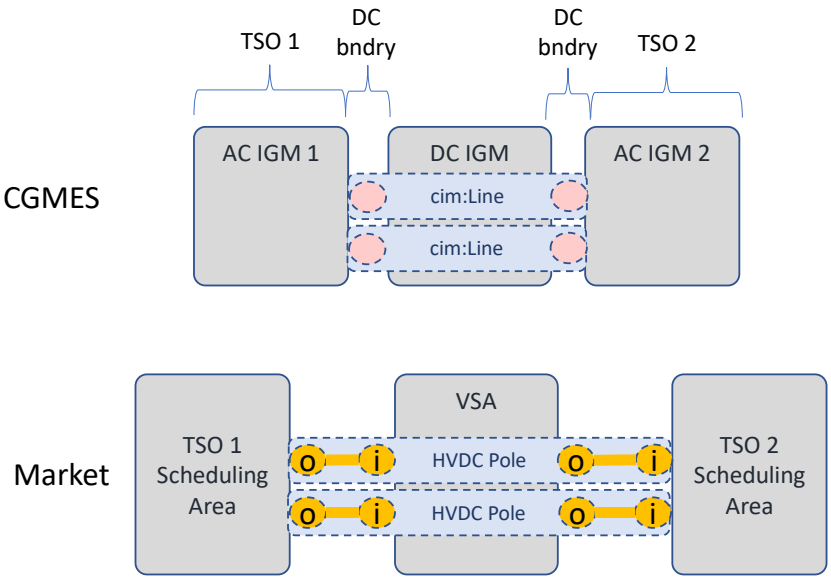


699

700 **Figure 15 HVDC boundary cases**

701 The upper part in Figure 15 shows the case with the DC IGM as the boundary and the lower
702 with the DC IGM internal to one TSO (TSO1) and both ends connected to the AC IGM1 managed
703 by that TSO. The lower part is also discussed in section 6.3.6

704 The market schedule concept manages the losses in the HVDC Link by a separate scheduling
705 area also denoted "virtual scheduling area" (VSA), the lost active power disappears in the VSA.
706 The market information model that match the upper part in Figure 15 is shown in Figure 16.



707

708 **Figure 16 Virtual Scheduling Area Market DC Interface**

709 The correspondence between market and CGMES concepts are shown in Figure 16;

- A TSO Scheduling Area in the market correspond to the cim:ControlArea in the AC IGM.
- The VSA do not exist in CGMES so there is no corresponding cim:ControlArea for it. A DC IGM do not have a cim:ControlArea as this is an AC network concept.
- A HVDC Pole in the market correspond to a cim:Line in CGMES (Option 1 in section 6.3.1) and the two cim:DCConverterUnits that have the EIC T code of the HVDC Pole.
- The boundary node in CGMES do not have a direct correspondence in the market.

The market schedule concept describes the power exchange between two scheduling areas by a time series. A time series describes the import and export between two scheduling areas. The exporting area is denoted “out_domain” and the importing area “in_domain”. In Figure 16 a time series is symbolized with a thick orange line with two circles at each end. The circles contain the letters “i” for “in_domain” and “o” for “out_domain”. As Figure 16 show a HVDC Bipole four time series are needed to describe the export from TSO 1 via VSA to TSO 2.

The lower part in Figure 15 describe a situation where the transfer over an internal HVDC Link appear at the AC boundary. The network in “AC IGM 1” between the HVDC Link and the AC boundary needs, in the general case, a power flow calculation to determine at which boundary nodes the active power is transferred.

6.4.2 Market schedule messages

This interface is only valid for HVDC Links between two scheduling areas. If a link is internal within a scheduling area refer to section 6.4.3.

Market schedules are defined per HVDC Link. A HVDC Link consists of one or more HVDC Poles where each HVDC Pole may be operated independently (examples on how poles are operated is given in section 7.3). Hence a schedule value for a HVDC Link needs to be divided on the HVDC Poles in operation.

According to the CGMA Business Requirements Specification (BRS) [10] the HVDC Link schedule is divided per HVDC Pole in the pole splitting and loss calculation (PSLC) process. This can be done either internally in the CGMA process or externally by a TSO or RSC. If done externally the TSO decide how to split the losses.

If done internally in the CGMA process, a simple model using parameters from the boundary definition is used, e.g. pole distribution factors and loss parameters in percent of active power transfer. The computed losses shall be added to the sending side so that the requested power from the receiving side is serviced.

If done externally by TSO or RSC they can benefit from the detailed HVDC model where a loss calculation formula describes how the components in a HVDC Pole contribute to the losses.

The pole split and loss calculation is required as HVDC Link schedules contains the total flow in all HVDC Poles without losses.

A description of the CGMA process is given in the CGMA BRS [10]. The inputs and outputs are

- Input pre-processing data (PPD) sent by each TSO. PPD contain schedules per HVDC Link and area interchanges. CGMA perform PSLC either internally or by getting the PSLC data from the TSOs, see next bullet.
- Input PSLC data sent by a TSO. The PSLC data contain schedules per HVDC Pole at sending and receiving side as well as the losses (difference between active power at the sending and

receiving side). PSLC data are provided only by TSOs connected to an HVDC Link and can be submitted by only one of the connected TSOs (single-side data provision).

- Output is the "final reference program" with area interchanges and HVDC Pole schedules aligned.

For post-market closure time frames, day-ahead and intraday, PSLC is always done externally by the TSO and provided as PEVF documents.

The above inputs and outputs are exchanged in XML documents with contents defined by the XML schemas "Reporting Information Market Document" (RIMD) and "Reporting Market Document" (RMD), both with the XML structure below

- Document root RIMD or RMD
  - sender_MarketParticipant.mRID, EIC V code of sending TSO in this case
  - receiver_MarketParticipant.mRID, EIC V code of receiving TSO in this case
  - subject_domain.mRID, describe the domain covered by the document. The element is used for several different things, e.g.
    - An HVDC Link.
    - An AC transmission line corridor.
  - TimeSeries
    - in_Domain.mRID that is the scheduling area receiving the power
    - out_Domain.mRID that is the scheduling area sending the power
    - businessType, the code "B68" indicate a HVDC Link schedule and the code "B63" indicate a HVDC Pole schedule.
    - connectedLine_RegisteredResource.mRID, gives the EIC-T code for the HVDC Link, businessType="B68", as well as the HVDC Pole, businessType="B63".
    - Period
      - Point
        - position
        - quantity

The RIMD/RMD document describe area interchange where the subject_domain.mRID defines a collection of HVDC Links, the in_Domain.mRID and out_Domain.mRID describe the scheduling areas, and connectedLine_RegisteredResource.mRID specify the specific link.

As the HVDC Link and HVDC Pole doesn't have a dedicated classes in CIM, e.g. cim:HVDCPole and cim:HVDCLink, the HVDC Pole is currently mapped to a boundary cim:Line.

The HVDC losses are taken by the sending side and this is accomplished by a virtual scheduling area (VSA) in the middle of the HVDC line as explained with the example in Figure 17.

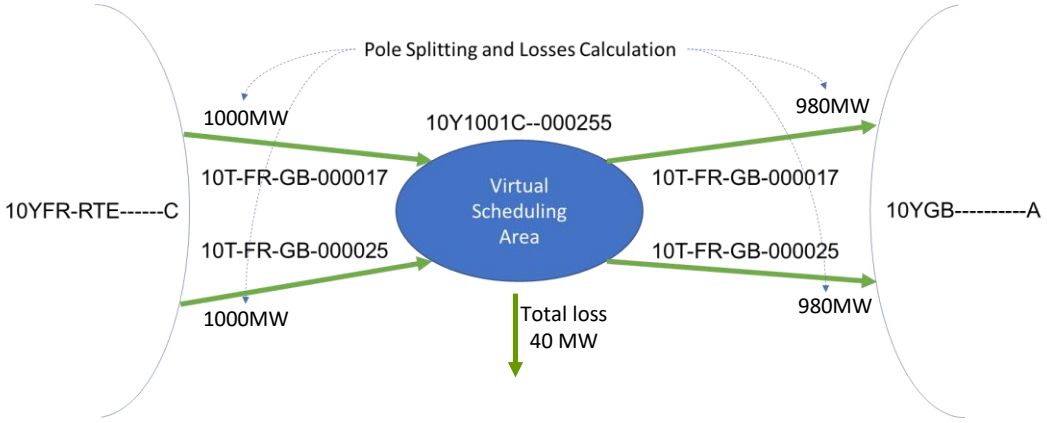


Figure 17 Example Pole Split and Loss Calcualtion (PSLC)

Figure 17 shows an example HVDC Link with two HVDC Poles (a bipole) identified by EIC codes 10T-FRGB-000017 and 10T-FRGB-000025 with bussinessType code "B63", see RIMD/RMD message description above. In the middle of the HVDC Link there is a virtual scheduling area cutting the HVDC Poles in two halves. In Figure 17 the loss is 2% of the transferred power and each HVDC Pole transfer 1000MW so the loss is 20 MW per HVDC Pole. The loss is added to the sending side, and is disappearing in the virtual scheduling area (VSA). In this example the losses are taken by the receiving TSO. To compensate for this the amount of transferred power can be increased in the HVDC Link schedule as agreed between the involved TSOs, e.g. if the agreement is that the losses shall be taken by the sending TSO this means that the sending TSO shall increase the transfer requested by the receiving TSO with the losses. For the example in Figure 17 assume that the receiving TSO requested a transfer of 1960 MW in total, the sending TSO will then have to increase the transfer with 40 MW to 2000 MW in total to reach the requested 1960 MW.

A RIMD/RMD message describing this PSLC contain the following four TimeSeries (schedules)

- Exchange from 10YFR-RTE------C to 10Y1001C—000255 (the VSA) using the HVDC Pole 10T-FRGB-000017.
- Exchange from 10YFR-RTE------C to 10Y1001C—000255 (the VSA) using the HVDC Pole 10T-FRGB-000025.
- Exchange from 10Y1001C------000255 (the VSA) to 10YGB----------A using the HVDC Pole 10T-FRGB-000017.
- Exchange from 10Y1001C------000255 (the VSA) to 10YGB----------A using the HVDC Pole 10T-FRGB-000025.

In the case of simplified HVDC the schedule values from the TimeSeries are picked up from the Period/Point/quantity at the time from md:Model.scenarioTime matching the Period/Point/position and are assigned to the EI.p attribute at the cim:EquivalentInjection connected to the boundary nodes at each side of the cim:Line as described in section 7.2.1. In case of detailed HVDC the assignment of schedules to the EI.p attribute is done for the sending side and the receiving side is computed by subtracting the losses from the sending side as described in section 7.2.1.

Two cim:Lines represents each HVDC Pole in the RIMD/RMD messages above. The sides of the

HVDC Pole is determined by the real scheduling area EIC code that match the cim:ControlArea with the in_Domain.mRID and out_Domain.mRID at the two sides.

A method that could be used to find the CGMES objects to update from the schedule is described in terms of CGMES objects and the schedule message below. But be aware that when the CIMXML files in IGMs are created a TSO internal model, not necessarily CGMES based, is used. The below method is described to show that sufficient data for it is present in CGMES.

- Each HVDC Pole is described by two TimeSeries in the schedule message, one for TSO1 to VSA and one for VSA to TSO2.
- The out_Domain.mRID and in_Domain.mRID in a TimeSeries refer to one of the following pair of EIC codes:
  - EIC_Y code of cim:ControlArea of TSO1 and EIC_Y code of the VSA.
  - EIC_Y code of cim:ControlArea of TSO2 and EIC_Y code of the VSA.
- The two TimeSeries that refer to the same VSA describe the flow from one TSO to the other.
- The EIC_Y code in the out_Domain.mRID define the area sending power and the EIC_Y code in the in_Domain.mRID define the area receiving power, the cases are
  - Export from TSO1 to TSO2
    - One TimeSeries have the out_Domain.mRID referring to the TSO1 cim:ControlArea and the in_Domain.mRID referring to the VSA.
    - The other TimeSeries have the out_Domain.mRID referring to the VSA and the in_Domain.mRID referring to the TSO2 cim:ControlArea.
  - Export from TSO2 to TSO1
    - One TimeSeries have the out_Domain.mRID referring to the TSO2 cim:ControlArea and the in_Domain.mRID referring to the VSA.
    - The other TimeSeries have the out_Domain.mRID referring to the VSA and the in_Domain.mRID referring to the TSO1 cim:ControlArea.
- For option 2 in section 6.3.1 the two cim:Lines, each containing one of two boundary nodes at the HVDC Pole, and the cim:DCConverterUnit at each side in the HVDC Pole is found using the EIC T code from the connectingLine_RegisteredResource.mRID attribute in the TimeSeries. For option 1 in section 6.3.1 the difference is that one cim:Line with two boundary nodes is found.
- The two cim:Line gives the two boundary nodes of the HVDC Pole.
- The AC IGM and the DC IGM (if exists) has a branch (e.g. cim:ACLineSegment) connecting it to the boundary node.
- The branch in the AC IGMs has a cim:Terminal that
  - connects to the boundary node in one of the the cim:Lines.

- has a cim:TieFlow that links to the cim:ControlArea. The cim:ControlArea gives the flow direction depending on the matching out_Domain.mRID or in_Domain.mRID as described above.

- The branch in the DC IGM has a cim:Terminal that connects to the cim:ACDCConverter with the cim:ACDCConverter.PccTerminal reference.

- The boundary node has an cim:EquivalentInjection both in the DC IGM and the AC IGM.

- The cim:EquivalentInjections and the cim:ACDCConverter are updated with the active power flow from the schedule.

6.4.3 Internal HVDC Link schedules

TSO internal HVDC Links transfer power within the TSO but may redistribute the power exchange over tie lines to other TSOs. In the case the HVDC Link is close to the border a single or a group of tie lines may carry the power transferred by the HVDC Link as described in section 6.3.6. In this case it is still the TSO with the HVDC Link that operates it and schedules the power transfer that is included in the power exchange schedule between the TSOs. The effect of the HVDC Link is then that it forces the transfer over the AC tie line to be the same or near the schedule for the HVDC Link. It is then important that the values from the schedule are properly transferred to the HVDC boundary injections (the EI.p attributes in SSH). If the HVDC boundary injections do not match the schedule values power flow may not be able to solve. For this reason, the internal HVDC Link schedules must be available for other TSOs and RSCs so consistency can be checked.

The market schedules described in section 6.4.2 are designed to describe the power exchange between scheduling areas which do not fit internal HVDC Link schedules where the power transfer on the link is scheduled and no scheduling areas are used. Instead TSOs have their own internal scheduling message formats and no commonly agreed standard exists for these messages. But a standard for TSO internal HVDC Link schedules is needed and such a proposal is included in Annex C.

7. Network data

7.1 Containment

This section describes the containment rules for HVDC equipment.

A converter is located in its own cim:Substation. The reasons for this are

- Converters are commonly built and organized in a separate converter substation.

- The boundary model does not support a boundary point within a cim:Substation, the boundary point need to be contained in a cim:Line. This is according to IEC TS 61970-600 and the more flexible boundary from IEC 61970-600 is not allowed, see also Table 32.

A converter substation includes

- One or more cim:DCConverterUnits and a cim:DCConverterUnit must be contained by a cim:Substation. The role cim:DCConverterUnit.Substation is optional in both IEC TS 61970-600 and IEC 61970-600 so a rule making it mandatory is needed.

- Optionally a cim:VoltageLevel where AC equipment is gathered, e.g. filters and shunts.

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A cim:DCConverterUnit can contain any AC objects, common objects are

- Subclasses of cim:ACDCConverter (cim:CsConverter or cim:VsConverter) that are both a AC and DC object.
- Subclasses of cim:Switch
- cim:PowerTransformer
- cim:ConnectivityNode and cim:TopologicalNode

Note that

- cim:VoltageLevels are not contained by cim:DCConverterUnit which means that AC objects in a cim:DCConverterUnit need a reference to cim:BaseVoltage.
- In general active and reactive sources as cim:SynchronousMachines, cim:StaticVarCompensators, shunts and filters shall not be located in the cim:DCConverterUnit but a cim:VoltageLevel.

The separation of data into DC and AC IGMs and the interface between them is described in section 6.

In the case filters are included in the converter substation they shall be in a cim:VoltageLevel as stipulated by existing containment rules.

The branches that connect with the boundary follow the containment rules, i.e.

- A cim:ACLineSegment is contained by the boundary cim:Line.
- A cim:Switch is contained by a cim:VoltageLevel or a cim:DCConverterUnit.
- A cim:PowerTransformer is contained by a cim:Substation or a cim:DCConverterUnit.

7.2 Levels of modelling detail

7.2.1 Overview

HVDC Poles can be modelled at several levels of detail depending on how much detail is available. The simplest model is a cim:EquivalentInjection (EI) with constant active/reactive power injections without further detail. For a voltage source converter the cim:EquivalentInjection (EI) can be constant active power injection and constant voltage which resembles a synchronous machine. The most detailed model includes the converters in the HVDC Pole.

The active power flow in the boundary cim:EquivalentInjection.p (EI.p) originate from pole splitting and loss calculation (PSLC) of HVDC Link schedules. In case of internal HVDC Links no market-schedules exists so the PSLC must be managed by the TSO having the HVDC Link.

In case there is a mismatch between the EI.p active power flow and the schedule values for an AC IGM using simplified HVDC, the EI.p is updated to match the schedule values.

If there is a mismatch between the EI.p active power flow and the schedule values for an AC IGM used with a detailed DC IGM, the EI.p values cannot be directly updated from the HVDC Pole schedule values. A power flow calculation will compute the HVDC Pole losses by using the loss formula from IEC TS 61970-600:2017 and the loss parameters from the DC IGM. The computed losses may differ from the losses used in the HVDC Pole schedules. Hence only the schedule values for the sending side shall be used to update the EI.p values at the rectifier

side. The EI.p values at the inverter side are computed by subtracting the computed losses from the schedule values at the sending side.

For a current source converter the corresponding EI.q may be computed using a formula describing the dependency between converter reactive power flow and active power transfer, see section 7.3.5.

The four levels of detail for modelling each side of a HVDC Pole, outlined in section 1.1, are described in the following sections.

7.2.2 Level 1 - Simplified HVDC using cim:EquivalentInjection without voltage control

Each side of a HVDC Pole is described by a cim:EquivalentInjection without voltage control in an AC IGM and without filters and shunts. Used with current source converters. This is a simplified HVDC model that cannot be combined with a DC IGM and shall only be used if the intent is to never assemble the AC IGM with a matching DC IGM.

No shunts and filters are available for voltage control. This alternative is described in AC IG [6] with additional detail added here. The cim:EquivalentInjection (EI) is a constant PQ source that doesn't reacts to voltage changes.

The reactive power in EI.q correspond to the reactive power consumed by the converter when the case was set up and changes in active power transfer due to contingencies is not reflected by a change in EI.q.

This option is intended for simplified HVDC only and it is not possible to assemble a detailed current source converter model from a DC IGM as the shunts and filters are not modelled in the AC IGM nor in the DC IGM and the EIs are not available to support the reactive power consumption for a converter in the assembled CGM.

7.2.3 Level 2 - Simplified HVDC using cim:EquivalentInjection with voltage control

Each side of a HVDC Pole is described by a cim:EquivalentInjection with voltage control in a AC IGM. Used with voltage source converters. This is a simplified HVDC model that can be assembled with a matching DC IGM.

The model uses simplified HVDC without shunts or filters. This alternative is described in AC IG [6] with additional detail added here. The cim:EquivalentInjection (EI) is a PV source that react to voltage changes by changing the reactive power injection. This gives a continuous response that do not match with the discrete response of the shunts and filters from a cim:CsConverter but describe the behaviour of a cim:VsConverter well. As the cim:EquivalentInjection also may have a reactive capability curve it may be used to describe the capability of a voltage source converter.

When this alternative is used at least the EI.minQ and EI.maxQ limits must be provided and the limit values shall match the reactive capability available at the active power flow specified in EI.p.

This option can be used with

- simplified model of current source converter where the shunts and filters are modelled as continuous reactive sources by the EI. Shunts and filters are not modelled in the AC IGM nor in the DC IGM. This model shall not be assembled with a DC IGM.
- simplified model of voltage source converter. This model can be assembled with a DC IGM.

7.2.4 Level 3 - Simplified HVDC using shunts and filters for voltage control

Each side of a HVDC Pole is described by a cim:EquivalentInjection without voltage control in an AC IGM and with shunts and filters in the AC IGM doing voltage control. Used with current source converters. This is a simplified HVDC model that can be assembled with a matching DC IGM.

This option includes shunts and filters not used for harmonics and voltage control. The cim:EquivalentInjection (EI) is a constant PQ source that doesn't react to voltage changes.

A change in the active power transfer is handled as follows

- EI.p is updated.
- EI.q may be updated to reflect the change in active power transfer, e.g. by using the simple formula Q=P/2 but according to Figure 30 Q=0.6*P is a better value. The choice shall be based on the specific converter characteristics and more detailed formulas are available, see also section 7.3.5.
- The filters used for harmonics may be updated to match the change in active power flow. But this can only be done if the function that describe the need for filters as a function of active power transfer is modelled.
- The shunts and filters not used for harmonics filtering are available for voltage control.

This option support DC IGMs with current source converters that are assembled into a CGM but can also be used without the DC IGM.

7.2.5 Level 4 - Detailed HVDC using shunts and filters for voltage

The HVDC Pole is described by assembling at least one of the AC IGMs with the DC IGM. This is a detailed HVDC model including one or two of the AC IGM(s) described in sections 7.2.3 and 7.2.4 above. The cim:EquivalentInjections at the boundary are not used and replaced by the the DC IGM. For a current source converter, the filters in the AC IGMs and are used for harmonics filtering. Filters not used for harmonics together with shunts or other equipment that can provide reactive power are used for voltage control. The DC IGM enables computing the dc voltages at the two converters and the dc current between them. The dc current is used to compute the losses in the DC Pole and this can only be done when the DC IGM is included.

7.3 CsConverter

7.3.1 CsConverter modelling

A current source converter consumes reactive power depending on active power transfer. When schedules are used to set up a power flow case only the active power is known so the corresponding reactive power consumption must be computed, and a formula is needed for this. The formula is also needed by power flow calculations to properly respond to contingencies or changes in active power transfer. An approximation is that the reactive power consumption is half of the active power transfer (Q=P/2), for more details see section 7.3.5.

Converters are commonly operated as monopoles or bipoles. A monopole typically has a single DC cable with ground return, but metallic return is used sometimes. Modern HVDC systems strive to keep ground return current low to minimize environmental impact which favors bipole solutions with two DC cables. In a bipole solution the current is circulating which means Idc1 = -Idc2 and that the ground return current is low, see Figure 18.

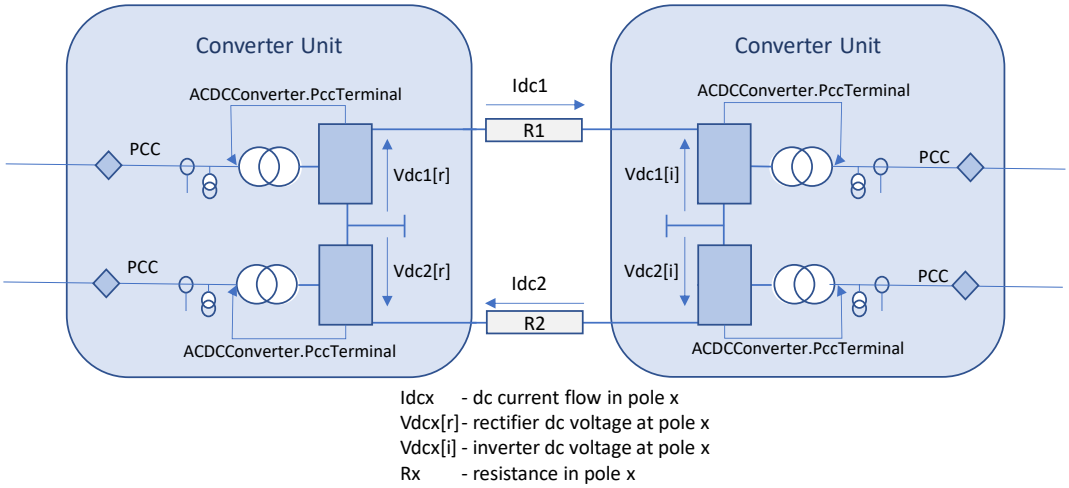


Figure 18 Example bipole configuration

In a HVDC Bipole as shown in Figure 18 has two HVDC Poles where one operates at positive voltage, the other negative and the midpoint between them is grounded. By keeping the Idc1 = -Idc2 the losses can be compensated for by slight difference in the voltages Vdc1[r] and Vdc2[r]. It is possible to operate the HVDC Poles in a bipole separately (happens in fault situations) but if ground currents are minimized this means that the HVDC Poles share the transferred power equally.

To minimize losses a CsConverter is typically operated at maximum dc voltage which reduce the dc current and the losses. A CsConverter also has a minimum power transfer capability (the attribute cim:ACDCConverter.minP is not available in IEC TS 61970-600:2017 but added in IEC TS 61970-600:2021). Below this limit the converter must be blocked, the limit is typically 2-10% of the max power capability.

The dc current in a HVDC Pole is determined by the voltage difference between the rectifier and inverter, i.e. Idcx = (Vdcx[r]-Vdcx[i])/Rx. To maintain the power flow from the rectifier to the inverter the inverter keeps a constant voltage Vdcx[i] and the rectifier controls the dc current by varying the voltage Vdcx[r]. Hence the cim:CsPpccControlKind seem to mix two levels of control as the cim:CsPpccControlKind.activePower relates to the HVDC Pole control while the cim:CsPpccControlKind.dcVoltage and cim:CsPpccControlKind.dcCurrent relates to cim:CsConverter control, see section 7.3.5 for a description of the control modes.

How the dc voltage depends on the ac voltage, tap position and firing/extinction angles is described by the formulas below.

$$U_{DC\,[rectifier\,]} = U_{DC0} * cos(\alpha) - \frac{3X_{tr}}{2\pi} I_{DC}$$

$$U_{DC\,[inverter\,]} = U_{DC0} * cos(\gamma) - \frac{3X_{tr}}{2\pi} I_{DC}$$

$$U_{DC0} = \frac{3\sqrt{2}}{\pi} * tapratio * U_{AC}$$

Where

- $U_{DC\,[xxxxxx\,]}$ is cim:ACDCConverter.udc at the rectifier and inverter.
- U_{DC0} is the rated dc voltage

- I_{DC} is the cim:ACDCConverter.idc.
- X_{tr} is the impedance at the transformer including any series reactance.
- α is the firing angle cim:ACDCConverter.alpha.
- γ is the extinction angle cim:ACDCConverter.gamma.
- $tapratio$ is the quotient between the AC voltages at each side of the transformer.
- U_{AC} is the voltage at the AC bus.

To be able to use these formulas the control function determining the firing/extinction angles and tap positions must also be known. The control function use the firing and extinction angle to change the dc voltage (Udc) and when the limits for these angles are reached the tap position is changed so the angles are moved to middle between the angle limits.

As discussed in section 7.3.5 the control functions depend on each other in a hierarchy where the firing/extinction angles and tap positions are decided at the lowest level. Before a lower level control function can be used the results from the control functions higher up in the hierarchy must be known e.g. the targets for dc voltages and the dc current. Section 7.3.5 suggests an algorithm for this control function. The firing/extinction angles impacts the harmonics generation and reactive power consumption. If the angle variation is big the angles must be known so reactive power consumption and harmonics generation can be determined. But if the angle variation is small and kept in tight limits the firing/extinction angles can be considered constant. Neither the tap position nor the angles are then needed in power flow calculations and it is assumed that the desired dc target voltages can be provided by the lower level control functions without the need to know tap positions nor the angles.

The parameters for the control of a HVDC Pole are described in cim:CsConverter and cim:VsConverter classes. Their base class cim:ACDCConverter has a reference, cim:ACDCConverter.PccTerminal, the location for determining the power flow (where the measurement transformers, CTs and PTs, are). This location is close to the PCC and is a cim:Terminal at a branch where the current is measured, hence the UML role name PccTerminal. The example in Figure 18 shows the point of common coupling (PCC), potential transformers (PTs) and current transformers (CT)s. The PccTerminal location is always on the line side of the converter transformer, not the converter side.

A current source converter bridge consists of six valves, see Figure 19.

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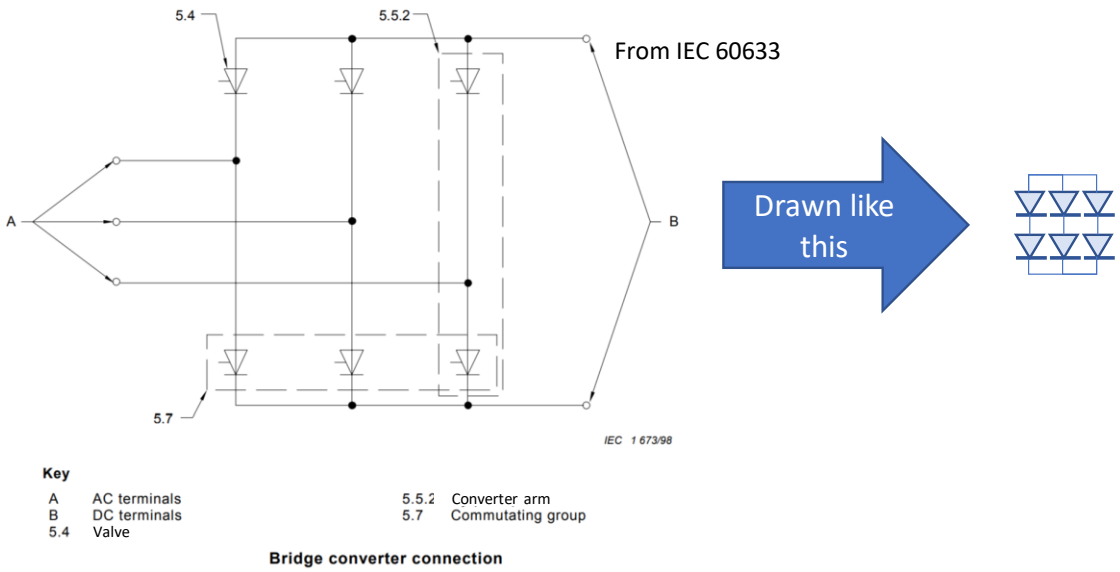


Figure 19 Current source converter bridge

The bridge generates 6 pulses with 60 degrees displacement per AC cycle. A monopole built with a 6-pulse bridge and ground return is shown in Figure 20.

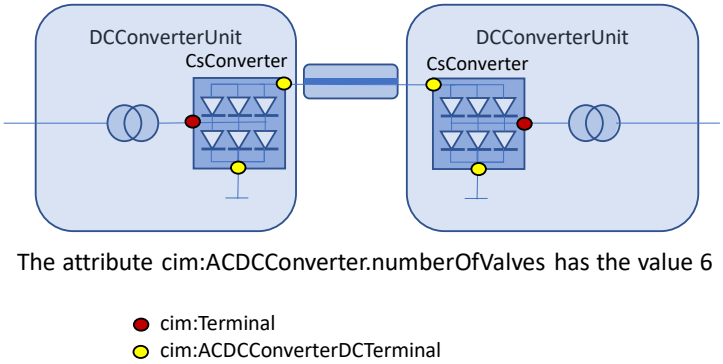
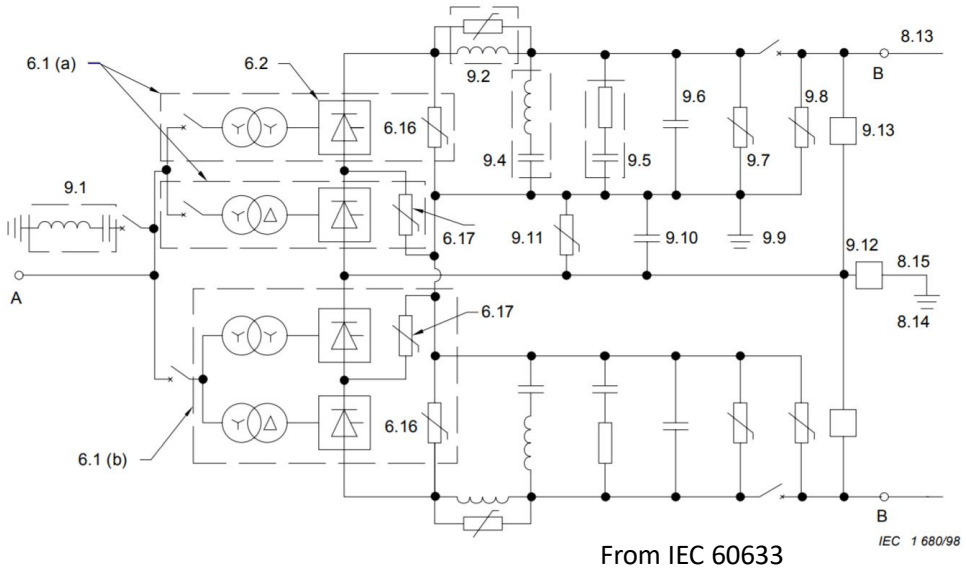


Figure 20 Example a monopole built with 6-pulse bridges

The monopole cim:CsConverters in Figure 20 has one cim:Terminal (red circle) connecting the transformer and one cim:DCTerminal connecting to ground and one cim:ACDCConverterDCTerminal connecting to the DC line (yellow circle).

The DC voltage generated by a 6-pulse bridge has lots of harmonics. A way to reduce them is to use two 6-pulse bridges to generate 12-pulses as shown in Figure 21.



From IEC 60633

Key

A	AC system	9.3	DC reactor arrester
B	DC terminal	9.4	DC filter
6.1 (a)	Converter unit (p = 6)	9.5	DC damping circuit
6.1 (b)	Converter unit (p = 12)	9.6	DC surge capacitor
6.2	Converter bridge	9.7	DC bus arrester
6.16	Converter unit d.c. bus arrester	9.8	DC line arrester
6.17	Midpoint d.c. bus arrester	9.9	Substation earth
8.13	HVDC transmission line pole	9.10	DC neutral bus surge capacitor
8.14	Earth electrode	9.11	DC neutral bus arrester
8.15	Earth electrode line	9.12	Metallic return transfer breaker (MRTB)
9.1	AC filter	9.13	Earth return transfer breaker (ERTB)
9.2	DC (smoothing) reactor		

Example of an HVDC substation

Figure 21 Example converter substation from IEC 60633

Figure 21 show two ways to configure the two 6-pulse bridges

- 6.1(a) with two separate converter units in series, each with one 6-pulse bridge.
- 6.1(b) with one converter unit having two 6-pulse bridges in series.

The lower transformer in 6.1(a) and 6.1(b) has a 30 degrees angle displacement relative to the upper one which results in 12 pulses per AC cycle making the DC voltage smoother and lower the cost for harmonics filtering. Figure 21 show two two-winding transformers but more common is one three winding transformer.

Configuration 6.1(a) in Figure 21 needs two cim:CsConverter objects which duplicate the number of control parameters. As the bridges are in series they cannot operate independently so one of the cim:CsConverter control variable sets is superfluous and shall not be used. Configuration 6.1(b) in Figure 21 needs one cim:CsConverter object which is a better fit with CIM. Hence configuration 6.1(a) shall not be used. A 12 pulse bridge shall not be modelled with two cim:CsConverters as would be the case for example 6.1(a), instead a single cim:CsConverter as shown in configuration 6.1(b) shall be used.

An example of a 12-pulse monopole modelled with a single cim:CsConverter is shown in Figure 22.

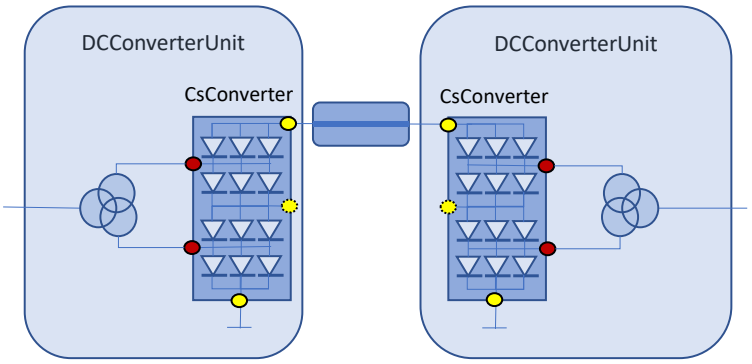


Figure 22 Example of a 12-pulse monopole

A single 12-pulse cim:CsConverter according to case 6.1(b) in Figure 21 has a single set of control/status variables and two cim:Terminals (red circles), one per bridge where each connects to one of the transformer terminals, one cim:ACDCConverterDCTerminal connected to the ground (in case of a monopole) and one cim:ACDCConverterDCTerminal connected to the DC cable (yellow circles). A third cim:ACDCConverterDCTerminal may exist between the two bridges which may happen if the point between the bridges is grounded (yellow dashed circles). A consequence is that a cim:CsConerter shall have one cim:Terminal in case of a 6-pules bridge and two cim:Terminals in case of a 12-pulse bridge. This can be avoided if a new class representing the bridge is added to CIM, the bridge has one cim:Terminal only.

The situation above appears because the cim:ACDCConverter class and its subclasses cim:CsConverter and cim:VsConverter describe the control function rather than the bridges. As most of the objects in a cim:DCConverterUnit are described by classes it would make sense to also add a class describing the bridge.

Valves in a currents source converter are thyristor based while voltage source converters use Insulated-Gate Bipolar Transistors (IGBT). The IGBT configurations are vastly different from what is shown in Figure 19 and may contain several levels of IGBTs.

Modelling the bridges in current source converters in CIM can be done using the following options

- (viable solution) Do not model the bridges with a new class as in the current solution. The cim:CsConverter and cim:VsConverter classes describe the control function and describe as many bridges as needed by a parameter.
  - (to be used) For cim:CsConverter the attribute cim:ACDCConverter.numberOfValves implies the bridge configuration. This solution is outlined above and correspond to alternative 6.1(b) in Figure 21, alternative 6.1(a) is not used for 12-pulse bridges. A 6-pulse bridge have one cim:Terminal while a 12-pulse bridge have two cim:Terminals.
  - (not to be used) For cim:VsConverter investigation is needed how cim:ACDCConverter.numberOfValves can describe IGBTs and their organization in levels. Multilevel VsConverters are out of scope until this has been investigated.

- (not to be used) Do not model the bridges with a new class as in the current solution and let cim:CsConverter describe one bridge. In a 12-pulse converter this means two cim:CsConverter objects of which one is a dummy where the attributes shall not be used. A 12-pulse monopole is modelled according to alternative 6.1(a) in Figure 21 with two 6-pulse cim:CsConverter objects. A bipole with two 12-pulse bridges is however modelled as two 12-pulse cim:CsConverter objects according to alternative 6.1(b) in Figure 21. These two ways of using the cim:CsConverter class to represent two different HVDC Pole configurations shall be avoided as it is confusing. This solution shall not be used.

- (future option) Add a new class cim:ThyristorBridge with six valves and an association with cim:CsConverter. The associations between cim:Terminal and cim:DCTerminal are moved from cim:CsConverter to cim:ThyristorBridge. What this would look like for a IGBT bridge is yet to explore. This requires a change in the CGMES 2.4.15 UML which means that this solution is out of the scope for CGM BP.

The phase reactor, that is often shown in circuit diagrams as figure 41 and 43 in IEC 61970-301:2020, is normally included in the transformer impedance but may also be represented by a separate cim:SeriesCompensator, especially in the case of a voltage source converter where the phase angle across phase reactor controls the active power flow.

7.3.2 Location of control objects

Shunts and filters may by design be located either in the AC substation (modern HVDC designs) or in the DC Substation (older HVDC designs), see examples in section 6.3. Filters and shunts may then be in the AC IGM while the converters are in the DC IGM. This is a consequence of splitting AC and DC models in separate IGMs.

Another consequence is that the voltage control object, e.g. cim:RegulatingControl, and the converter may be in different IGMs (see example in Figure 7 and section 6.3). If parameters from the converter are needed by the control function a path between the control object and the converter is needed. A direct reference between the control objects in the AC and DC IGMs is not possible as the reference will be dangling until inclusion in a CGM. A solution is to use the cim:Terminals at the branches at each side of the boundary node as controlled cim:Terminals. The control object can then be linked to the converter via the boundary node and the roles cim:RegulatingControl.Terminal and cim:ACDCConverter.pccTerminal. If the cim:ACDCConverter.pccTerminal or cim:RegulatingControl.Terminal is not referring to a cim:Terminal at the boundary branch, a topology search between the controlled cim:Terminal and the boundary cim:Terminal is needed. Note that TSO internal detailed HVDC models are also exchanged as a separate DC IGM so a branch cim:Terminal is present also in this case.

This is a complex and fragile solution that can be avoided as the control function can be described as a function of transferred active power only, e.g. the functions for filtering of harmonics and converter reactive power consumption. This means that the function for activating/deactivating filters and the voltage control can be solved in the AC IGM without a need for the DC IGM.

7.3.3 Transformer control

This section discusses the details of controlling the converter transformer tap. This control is related to the firing and extinction angles as discussed in section 7.3.5. As these angles reach their limits the tap is adjusted which brings the angles back into the middle of their ranges. For power flow calculation this level of detail is not needed and the discussion in this section need not be considered for power flow but a dynamics study the tap position and angles are

needed.

The power transformer in a cim:DCConverterUnit has no association with the converter that control its tap changer. The tap changer position determines the dc voltage and hence the active power transfer, an association that links them together is needed to resemble how all other controls are related but this require CIM/CGMES changes.

Power transformers in the AC system can be used to control voltage (in phase) and/or phase shift using tap changers. Such transformer models contain the following CIM objects

- One cim:PowerTransformer.
- Two or three cim:PowerTransformerEnds.
- At max two cim:TapChangers per cim:PowerTransformerEnd.
- Zero or one cim:TapChangerControl per cim:TapChanger.
- Each cim:TapChangerControl must have a reference to the controlled point being a cim:Terminal.

Power transformers in a cim:DCConverterUnit are used to control the voltage at the converter bridge which impacts the DC voltage at the DC side of the converter bridge. Assuming constant voltage at the AC side there are two methods to control the DC voltage, moving the transformer tap changer or change the firing or extinction angles. The HVDC control system uses both to control the power flow over the link.

To describe a power transformer in HVDC the following objects are needed

- One cim:PowerTransformer that describe the electrical connectivity.
- Two or three cim:PowerTransformerEnds that are needed to describe the transformer reactance's.
- One cim:RatioTapChanger that describes the tap changer.
- A cim:TapChangerControl shall not be used to control the tap changer. Instead a cim:CsConverter or cim:VsConverter describes control of the tap changer.

The cim:CsConverter or cim:VsConverter doesn't have a direct reference to the controlled cim:RatioTapChanger which means that a search starting from the converter to find the cim:PowerTransformer with the cim:RatioTapChanger is needed. To avoid this an association between the cim:RatioTapChanger and the cim:ACDCConverter need to be added. This means a change of the CGMES/CIM information model is out of scope of this document.

A temporary solution is to use the containment where the power transformer and converter are contained by the same cim:DCConverterUnit and use this to navigate from the converter via the transformer to the tap changer. For this to work the containment rules must be followed, e.g. each converter shall be contained by its own cim:DCConverterUnit.

7.3.4 HVDC harmonics filtering, reactive compensation, and voltage control

Harmonics appears at specific multiples of the base frequency, e.g. 11, 13, 21, 23 etc. The current caused by the harmonics at the multiples depend on active power transfer. The higher harmonics disturbance the more filters are needed. The filters also provide reactive power that compensate the reactive power consumption by the converter. Filters can be designed to

remove all harmonics or be dedicated to specific harmonics multiples. Filter design also impact the amount of reactive power generated. This must be considered in contingency studies so the power flow calculation can respond to contingencies by proper use of filters and shunts, i.e.

- At the level of harmonics disturbance determine which filters to use. The remaining filters are available for voltage control.
- Use the shunts and filters not used for harmonics filtering or other available reactive power sources to maintain the voltage. The voltage control function will also compensate the reactive power consumed by the converter.

According to section 7.1 the detailed model of the converters are located in their own substation that is connected with the AC substation. Section 7.1 also describes ways of having AC equipment in the converter substation which means that the shunts can be in the

- Converter substation as is the case with older current source converters. In this case the substation can be split as described in section 6.3.5, see example in Figure 23.
- AC substation as is the case with modern current source converters, see example in Figure 24 below.

Figure 23 shows an example of split converter substation, it is the same as in Figure 11. A fictitious substation is created in the AC IGM where the shunts and filters are in a cim:VoltageLevel in the fictitious converter substation. A cable connects it with the AC substation.

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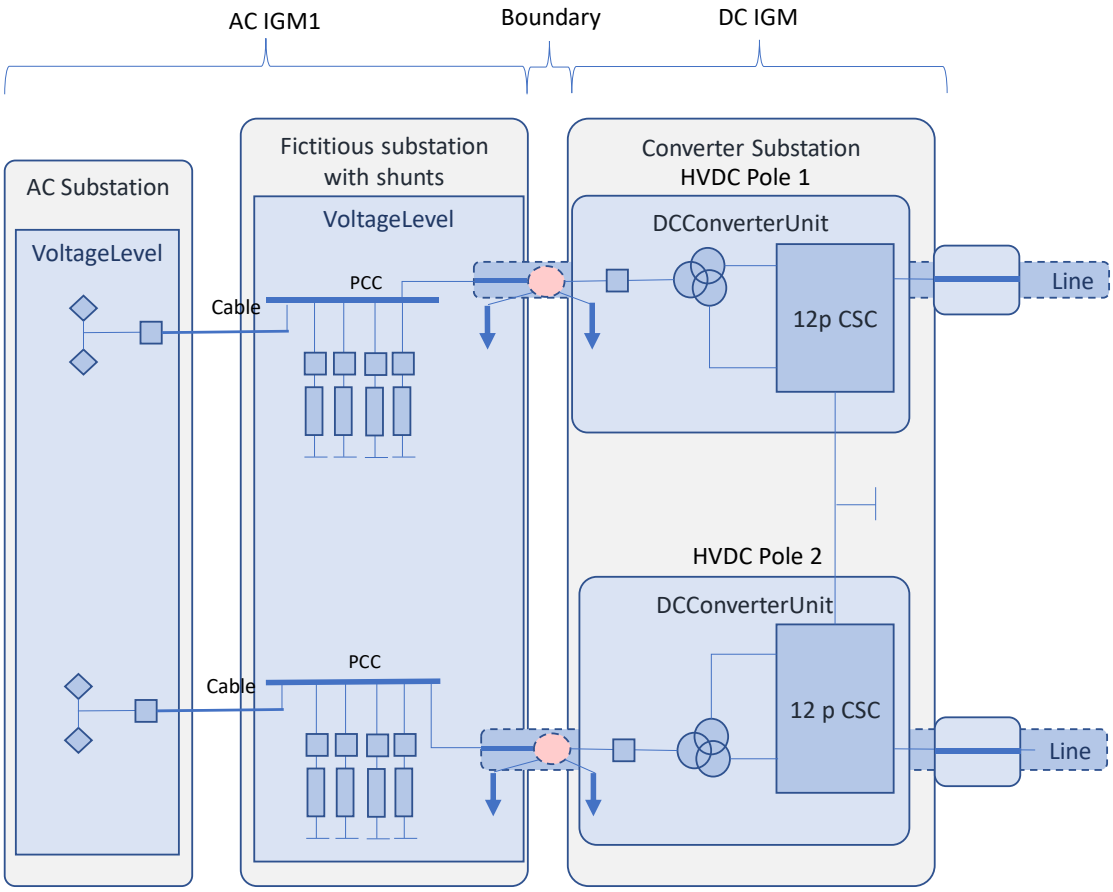


Figure 23 Example with filters and shunts in split converter substation

Figure 24 shows an example with the shunts located at two different bus bars in the AC substation, other similar example are shown in Figure 9 and Figure 11.

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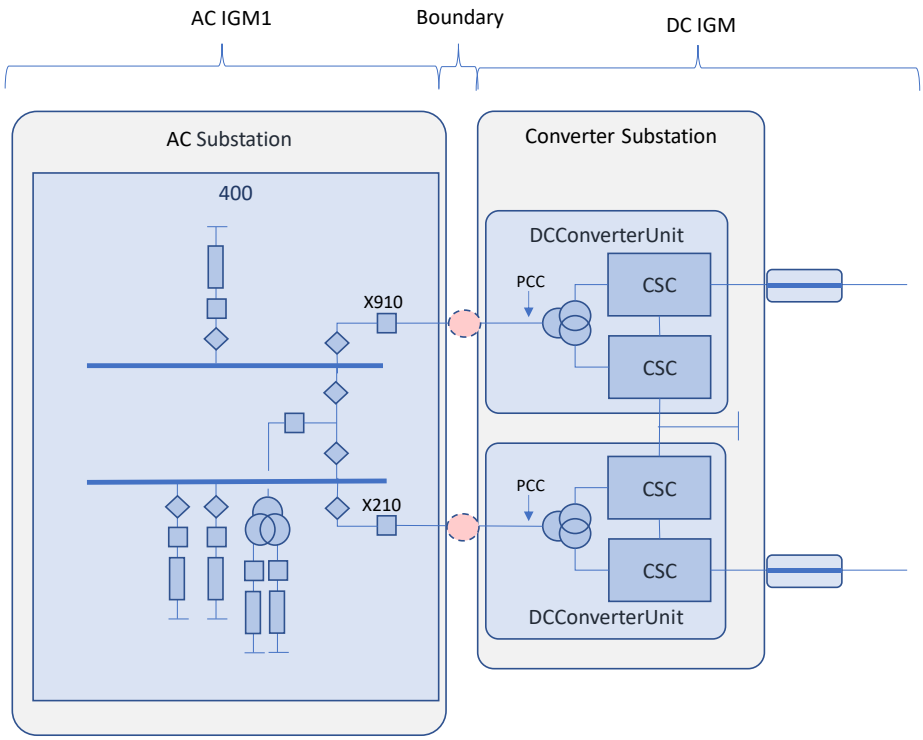


Figure 24 Example with shunts in the AC substation

The examples in Figure 9, Figure 11, Figure 23 and Figure 24 has the advantage that the shunts and filters are available without the DC IGM and can be used for realistic voltage control.

In either case above the control object need to be in the IGM where the shunts are not to avoid dangling references (cim:RegulatingCodEq.RegulatingControl) as discussed in section 7.3.2. For the examples in Figure 23 and Figure 24 the control object shall be in the AC IGM. The example in Figure 12 shows the same substation as in Figure 23 but without splitting the converter substation. The converter substation is in the DC IGM and the control object must also be in the DC IGM to avoid dangling references.

The filters used for harmonics are determined based on the active power flow. Depending on filter design they will also compensate the converter reactive power consumption. Any mismatch is taken care of by regular voltage control using shunts and filters not used for harmonics filtering.

A current source converter generates harmonics at multiples of the base frequency. The converter design determines at what multiples harmonics appear. Filters are typically designed the remove harmonics at specific multiples, e.g. 11th and 13th. Figure 25, from an IEEE paper, show an example filter design for the Three Gorges project in China.

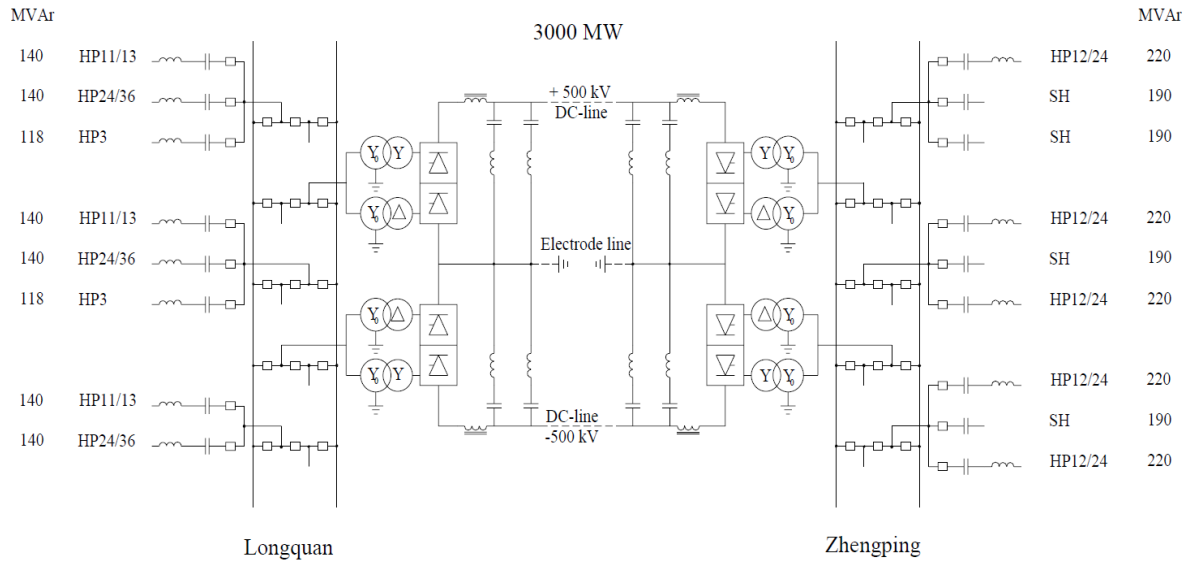


Figure 25 Example filter configuration from Three Gorges China

Figure 25 show a 12-pulse converter bipole that can be flexibly connected to the filters and shunts via a bus bar system at each side. The very left and right columns shows the reactive power generated by a filter when connected and next column to the middle show the harmonics multiples taken care of by the filter, e.g. HP11/13 take care of the multiples 11th and 13th. There are also several filters tuned for the same harmonics multiples that are connected based on need.

The reason why a substation is built with the flexible configuration options exemplified in Figure 25 is to support recovery from component failures. Normally all equipment needed is connected and the substation is not split into two buses operated separately.

Figure 26, from a set of trainings slides on harmonics, show an example on how harmonics generation depends on active power transfer.

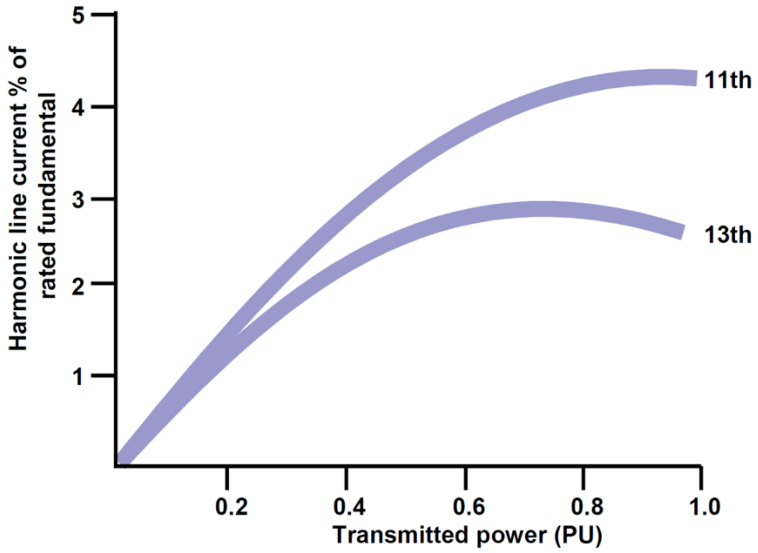


Figure 26 Example of 11th and 13th harmonics multiples as function of active power transfer

Figure 26 show how the harmonics current increase with increased active power transfer and that the dependency is nonlinear, other examples show that the curves may have quite

different shapes. The thresholds for connection or disconnecting filters are set by a quality index for acceptable level of harmonics currents. The harmonics generation, filter design and control of the filters is highly dependent on the installation and it is difficult to come up with a general analytical expression for control of the filters. A commonly used solution is to describe the filter control by a table where the activation and deactivation of filters is a function of active power transfer. To implement this in CIM will need a new CIM class that describe tabular functions. The filter control function will also need a new CIM class that manage connection and disconnection of filters using the result from the look up in the tabular function. The filters used for harmonics filtering must be excluded from voltage control and this can be accomplished by setting the shunt cim:RegulatingCondEq.controlEnabled flags to "false". This is managed by the filter control function. Figure 27 shows an example filter control function as a table.

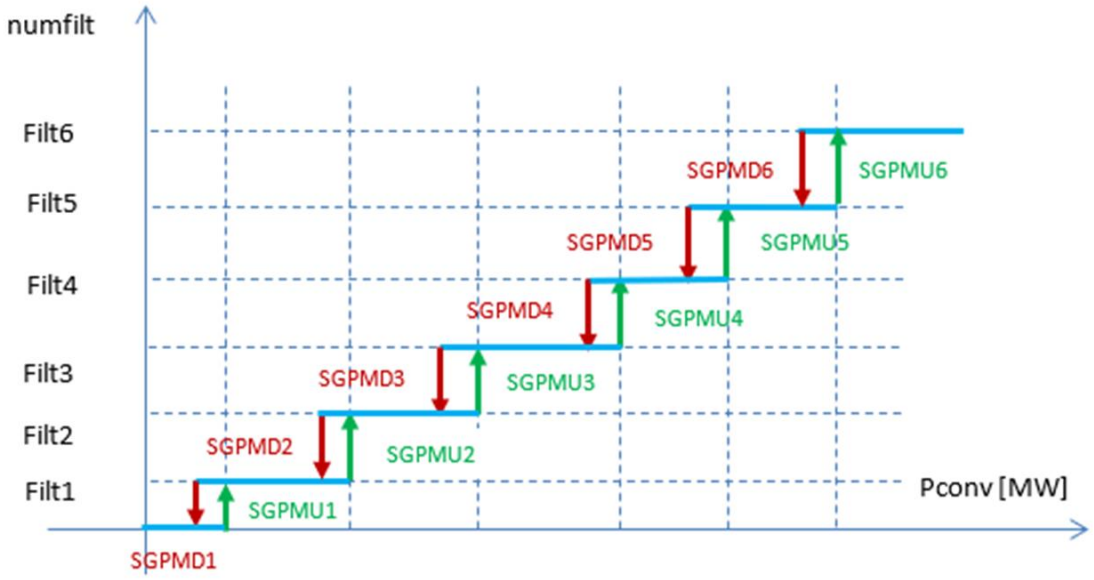


Figure 27 Example filter control function

To avoid extensive switching of the filters hysteresis is needed so that the switch off threshold is lower than the switch on threshold. The red arrows in Figure 27 show the switch off thresholds and the green the switch on thresholds.

If the filter control function depends on active power transfer only it does not need to be with the converter and can instead be linked with the cim:RegulatingControl controlling the voltage at the AC side. It is then easy for the filter control function to reserve and release the filters by changing the cim:RegulatingCondEq.controlEnabled flags. On the contrary, if the filter control function is located at the DC side it needs a path to the cim:RegulatingControl using filters for voltage control. The path will cross the boundary and given that cim:ACDConverter.PccTerminal and cim:RegulatingControl.Terminal refer to cim:Terminals at the branch ends connected to the boundary node the path is cim:ACDConverter.PccTerminal – BoundaryNode - cim:RegulatingControl.Terminal. This depends the restrictions on the cim:ACDConverter.PccTerminal and cim:RegulatingControl.Terminal which makes the solution fragile. The best solution is if the filter control function can be placed at the AC side and be independent of the converter.

Conclusion is that filter control with CIM is not possible until CIM has been extended to support it.

7.3.5 CsConverter control

HVDC Link control is divided in levels, disregarding the power system control the levels are

1. HVDC Link control
2. HVDC Bipole control. The two HVDC Poles in the bipole are coordinated at this level and the active power exchange over the two HVDC Poles is controlled.
3. HVDC Pole control. The target is active power transferred over the pole which is controlled as follows
   c) The converter with the lowest voltage determines the dc Voltage, the converter operates in voltage control mode, this is the inverter.
   d) The converter with the highest voltage determines the dc Current according to the formula Idc = (Vdc[r]-Vdc[i])/R, see Figure 18. The rectifier operates in current control mode and the inverter in voltage control mode.
4. HVDC Converter unit control. Input is dc Voltage or dc Current target.
   e) In voltage control mode the dc Voltage is controlled by the tap changer and firing/extinction angels. The angle control is fast but has a limited range, see Figure 30. Hence the angles are used track changes and when the limit is reached the tap position is changed. The target firing/extinction angels and tap position are the output.
   f) In current control mode the dc Current is controlled by setting the dc Voltage at the sending side and voltage control is used to accomplish this.
5. Bridge control, that given the target angle, generates the ignition pulses to the thyristors.

Figure 28 shows the control hierarchy levels 2, 3 and 4 described above.

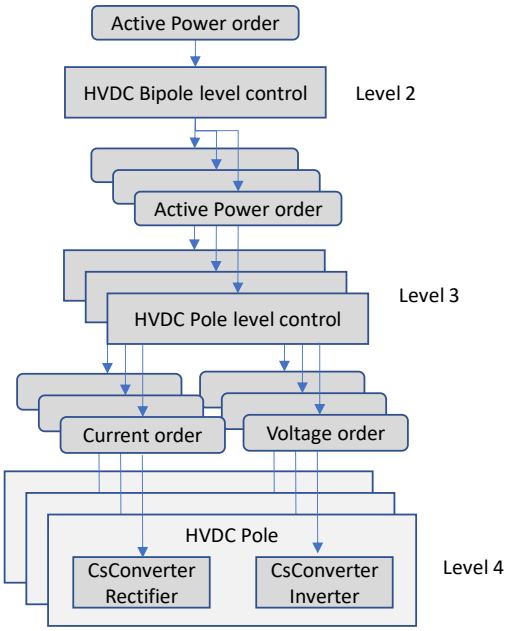


Figure 28 CSConverter control hierarchy

In the case that the active power flow needs to be rescheduled, e.g. due to a contingency or remedial action, the reaction of the above control functions, need to be described. A change at the upper levels will ripple down to the lower.

Level 1 in Figure 28 is the HVDC Link scheduling process that determine the flow for individual HVDC Bipoles and HVDC Poles.

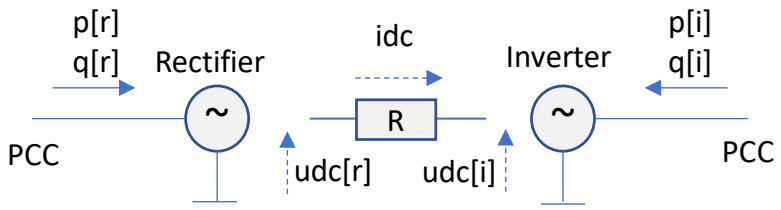
Level 2 in Figure 28, the input for CGM BP comes from the pole split and loss calculation function, see section 6.4. If the power flow is changed due to a remedial action or outage the change of the power flow needs to be described so it can be implemented in power flow. The cases are

- Loss of a HVDC Pole in a bipole. The remaining HVDC Pole stay at the target.
- Change of active power transfer due to remedial action, the change is shared evenly by the HVDC Poles in a HVDC Link regardless of monopole of bipole.

Level 3 in Figure 28 takes the active power transfer as target. The targets are set in the El.p attributes at the AC/DC boundary. Using simplified formulas to determine reactive power consumption, harmonics filtering and loss calculation the dc voltages and dc currents need not be computed. But if the losses are to be computed the dc current and dc voltage is needed. In the case of a remedial action the active power targets are changed which means that the dc voltages and currents must be recomputed from scratch.

Level 4 in Figure 28 takes the target dc Voltage or dc Current as input and generate the tap position and firing/extinction angels as output. The harmonics is heavily dependent on the firing/extinction angles as shown in Figure 30 and if they are not kept in a narrow band (which they are) this will impact the filter control. If the angles vary a lot, the control function that determine the angles and the tap position is needed. Support for this is needed in the CIM/CGMES specifications and due to this the function is out of the scope for this specification.

Power flow calculations need to support level 3 but not level 4. Figure 29 show an equivalent model for this.



$p[r] = udc[r]*idc$

$p[i] = -(p[r] - poleLossP)$

$poleLossP = idleLoss + switchingLoss*|idc| + resitiveLoss* idc^2$

Figure 29 Modelling the CsConverter in power flow

The injected power at the rectifier is P1 = udc1*idc. To compute the power delivered by the inverter we need to subtract all the losses in the HVDC Pole from P1 as showed by the second equation in Figure 29. The losses (Ploss) are a function of Idc as shown by the third equation in Figure 29. To compute Idc we can decide on a value for Udc1. As the losses increase with the square of the current a high value will minimize the losses. For this reason the rated voltage (cim:ACDCConverter.ratedUdc) is used.

An algorithm to compute the transferred power p[i] from the inverter at a given active power sent from the rectifier is

- The active power p[r] at the rectifier is targetPpcc[pole] from the level 2, in this case a HVDC Bipole. The value is picked up from the HVDC Pole schedules for this boundary node.

- Compute the dc current at the rated voltage at given active power transfer at the rectifier
  - idc = p[r]/ratedUdc
- Compute the losses at this current
  - poleLossP = idleLoss + switchingLoss* | idc| + resitiveLoss* idc^2
- Compute the power delivered by the inverter as p[i] = -(p[r] – poleLossP). This is the value at EI.p at the inverter.

All parameters above are from the cim:ACDConverter, cim:CsConverter or cim:EquivalentInjection.p (EI.p) if nothing else indicated. The resistiveLoss attribute also contains the equivalent losses from both transformers as well as any ground resistance.

The above algorithm put all the losses at the inverter side assuming that the HVDC Pole schedules already considered how the losses are split

Shunt capacity is used keep cos(ϕ) = 1 so that no reactive power is pulled from the AC network. The harmonics filtering has priority over getting cos(ϕ) = 1 which could result in production of more reactive power than consumed by the converter which tend to raise or lower the voltage. Over voltages are reduced by use of shunt reactors or static var compensators.

The reactive power consumed by a current source converter is roughly 45 to 65 percent of the transferred active power. The curve in Figure 30 show how the reactive power consumption typically depends on the active power transfer at fix firing and extinction angles.

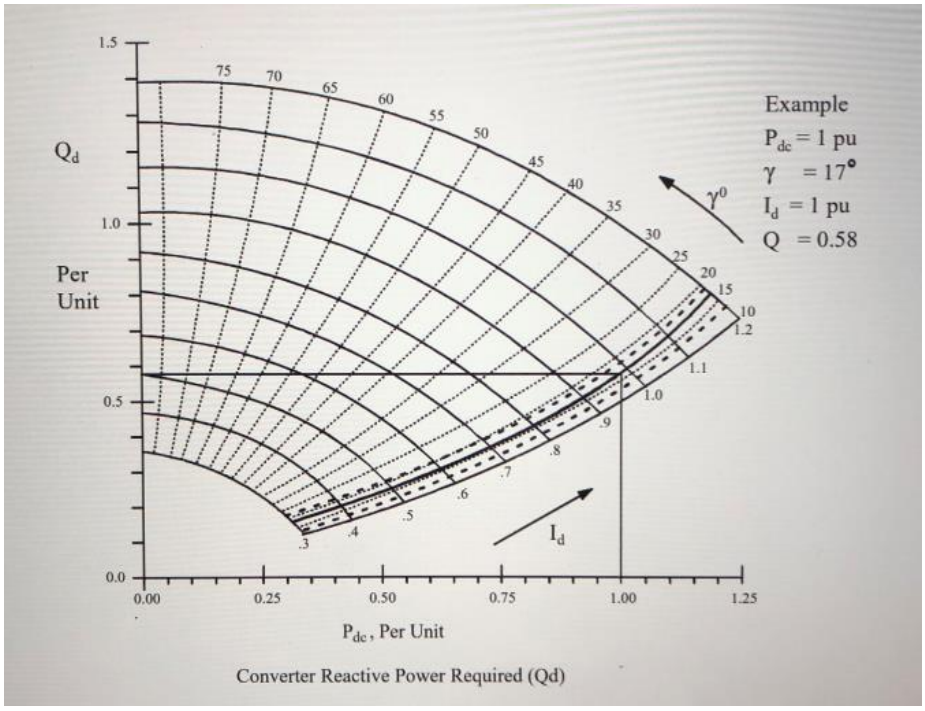


Figure 30 Reactive power consumption as function of transferred DC power

Figure 30 is from reference [13] show how the reactive power consumption increase more

than linearly with increasing power transfer due to increased current lag resulting in decreasing power factor cos(ϕ). The reactive power consumption is also dependent on the firing and extinction angles as shown in Figure 30. Both angles are kept in a small range which avoids big changes in reactive power consumption due to angle variations, a typical range for the extinction angle is indicated by the dashed lines in Figure 30. Due to this the shunt control needs to switch capacitors or reactors in or out to provide reactive power matching the reactive power consumption, this is regular voltage control. The reactive power consumption by a converter is described by a function Q = q(P). Detailed formulas for q(P) can be found in IEC 61970-301:2020 (Edition 7) and in HVDC literature, e.g. [13]. Assuming that the firing/extinction angles is considered fixed a commonly used simplified formula is Q=P/2 or Q=0.6*P. A better approximation is Q=P*sqrt((udc/udc0)^2 -1) where udc is the dc voltage and udc0 is the rated dc voltage.

For a business process where the active power transfer is known but not the reactive power consumption (e.g. 1D and 2D) a formula for computing it is useful.

7.4 VsConverter

7.4.1 VsConverter modelling

The voltage source converters differ from the current source converters in that both active and reactive power transfer can be controlled independently. The active and reactive power must stay inside a reactive capability curve not to overload the converter. An example reactive capability curve is shown in Figure 31.

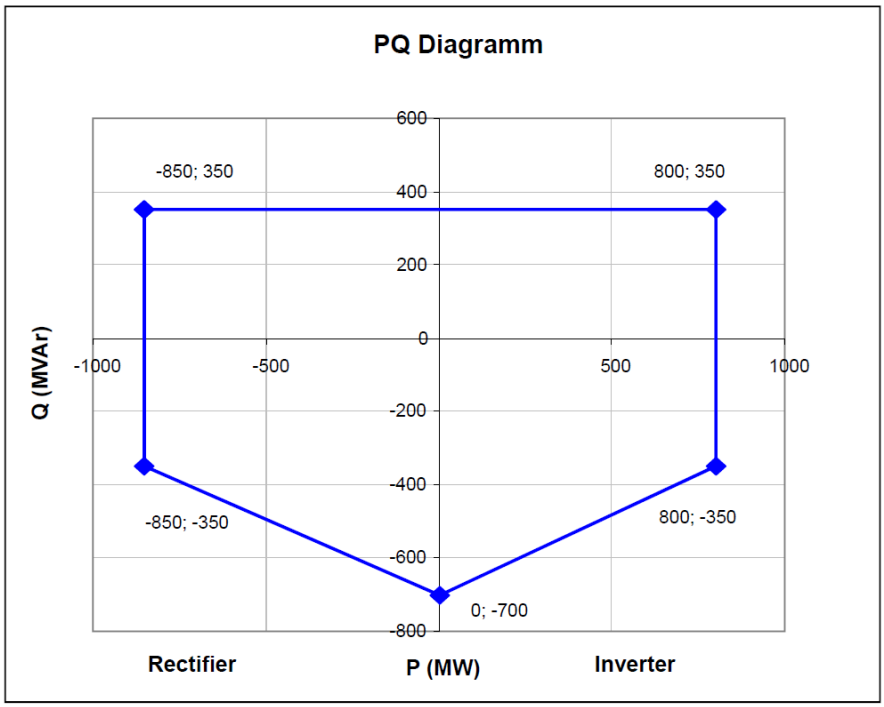


Figure 31 Example reactive capability curve for a VSC

Within the curve, the operation is continuous so no minimum active power limit around zero power is needed as for the CsConverter.

The basic components of a voltage source converter are showed in Figure 32.

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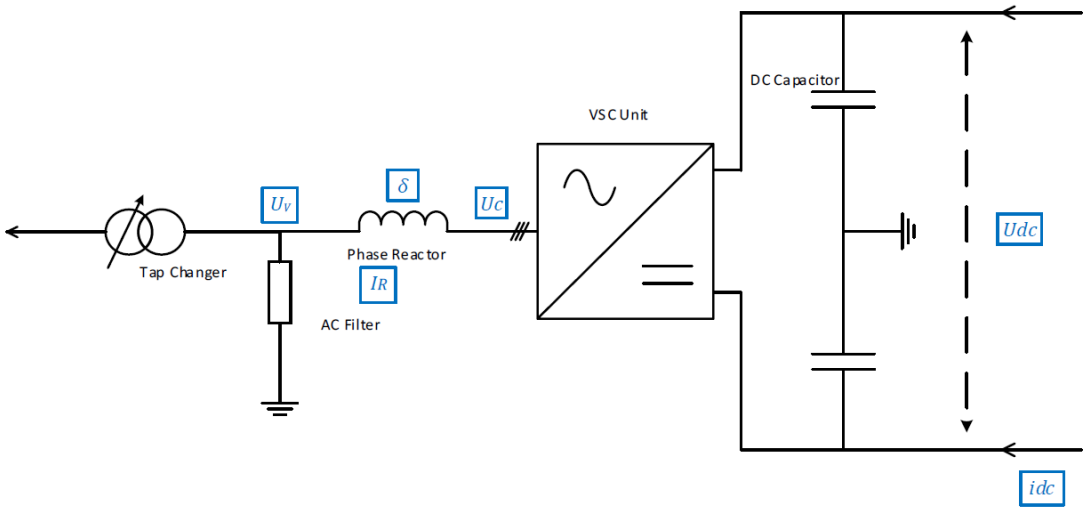


Figure 32 Voltage source converter

Voltage source converters are built with multiple Insulated-Gate Bipolar Transistor (IGBT) that are organized in multiple levels. The VSC Unit in Figure 32 is typically built with IGBTs. Each IGBT can be turned off and on which means that they do not depend on the AC voltage for commutation. In modern voltage source converters, the IGBTs are turned on and off at high frequency called pulse width modulation (PWM). PWM together with multiple switching modules (SMs) in series creates an almost harmonics free AC current which result in inexpensive filter solutions.

The active power flow in the converter can be described by the angle between the voltages Uv and Uc while the reactive power flow can be described by the magnitude difference between Uv and Uc, the formulas are

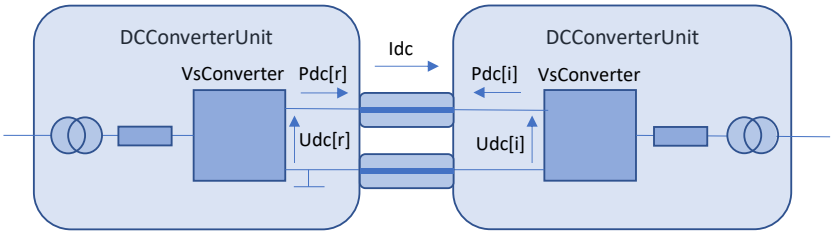
- P = Udc*Idc = Uv*Uc*sin(δ)/Xc
- Q = Uv*(Uv-Uc*cos(δ))/Xc

The variables in the formulas are shown in Figure 32. Additionally, δ is the angle between Uv and Uc. Xc is the Phase reactor reactance, see also IEC 61970-301:2020 for details.

For a VsConverter the tap changer does not have the same active role in controlling the power transfer as in a CsConverter which means that voltage Uv is stable. This means the angle δ and amplitude of Uc are the two controlled entities. The quotient Uc and Udc amplitudes, called the modulation index, is used to control the Uc amplitude and the reactive power flow.

As the VsConverter active and reactive power control is continuous and decoupled it is simpler to describe for use in power flow calculations than for a CsConverter where converter reactive power consumption and harmonics filtering must be described.

Figure 33 show a complete HVDC Pole for a monopole VsConverter (VSC) with metallic ground return. The filters in VSCs do not provide the amount of reactive power as for a CsConverter, hence they may be excluded.



Idc	the dc current
Udc[r]	the dc voltage at sending side
Udc[i]	the dc voltage at receiving side
Pdc[r]	the active power at sending side
Pdc[i]	the active power at receiving side

Figure 33 Control of a monopole VSC with metallic ground return

The active power flow across the DCLineSegment follow the same equation as for the CsConverter, relating to Figure 33

- Idc = (Udc[r] – Udc[i])/R where R = cim:DCLinesegment.resistance.
- Pdc[1] = Udc[r]*Idc = Udc[r]* (Udc[r]– Udc[i])/R

The side sending active power is referred to as the "rectifier" indexed [r] and the side receiving active power as the "inverter" indexed [i].

The sides operate like a CsConverter where

- The sending side, in Figure 33, determines the active power flow by controlling the difference Udc[r] – Udc[i]. As Udc[i] is not available the angle δ, that is measured, is used instead.
- The receiving side determines the voltage Udc[i].

Each side control their own reactive power exchange with the AC system.

There are numerous ways to create the PWM pulses and how this is done is outside the scope of this document.

The losses are computed the same formula as for CsConverters

- poleLossP[1] = idleLoss + switchingLoss* |Idc| + resistiveLoss* Idc^2

IEC 61970-301:2020 has a formula with an additional linear resistiveLoss term which may be an error.

7.4.2 VsConverter control

The PWM pulses are generated according to a hierarchical scheme like the CsConverter. The levels for the VsConverter leaving the upper levels out are

- Converter control
  - One side control the active power with the targetPpcc as input and its own reactive power using targetUpcc for voltage control or targetQpcc for reactive flow control at the PCC.
  - The other side control its dc side voltage with the targetUdc and its own reactive power using targetUpcc for voltage control or targetQpcc for reactive flow control at the PCC.

- Switching Module control generates the PWM pulses with inputs from the converter level control. The details for this are out of the scope for this document.

In the same way as for a synchronous machine it can be assumed that that active and reactive power can be controlled independently. As there is no need to consider harmonics nor reactive power compensation similar power flow modelling as for a synchronous machine can be used.

The active power can be managed in several ways

- Scheduled power where the active power injections are fixed according to a schedule. As the HVDC Link do not participate in load frequency control the link do not participate in keeping the frequency stable.

- Variable impedance where the transferred active power corresponds to the angle difference across the HVDC Link. The HVDC Link exhibit a series impedance and transfer power as a normal cim:ACLineSegment with the difference that the impedance value can be varied. The current CIM do not support this case and it is not in scope of this document but a note to support it in future versions of CIM has been added in Annex A.

- Frequency control in an isolated area connected via an HVDC Link to a main network.

The same equations as showed in Figure 29 applies with exception for the loss formula above that include an additional linear loss term for the dc current. The power flow model is the same as in section 7.3.5. Note that all controls relate to PCC and remote control is not possible.

The reactive power injected by a synchronous machine is a step function where the reactive power injection is a function of the voltage, see Figure 34

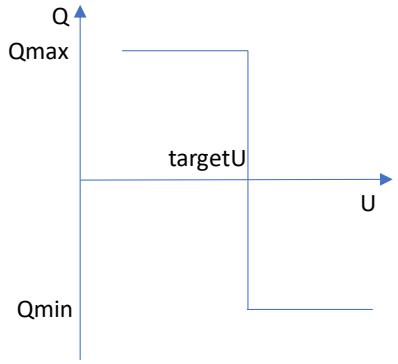


Figure 34 Typical Synchronous Machine Voltage control in power flow

The Qmax and Qmin limits are either from the two fixed limits or from the reactive capability curve. The reactive power injection is solved by the power flow and stays at the vertical line in Figure 34. If a limit is reached the power flow bus is changed to PQ type and the voltage is computed instead. In case multiple machines are controlling the voltage in parallel the reactive net injection is divided on the machine according to some ad hoc method, e.g. by sharing according to the reactive range.

A more accurate model is to describe the response to a voltage change with a droop curve, see Figure 35.

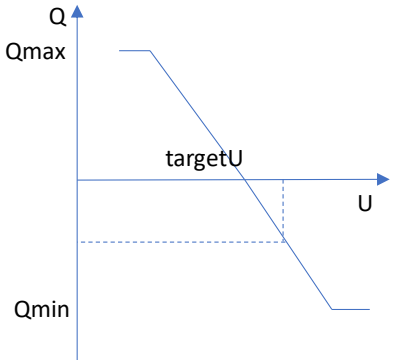


Figure 35 Voltage control droop curve

As the voltage change, see example dashed line in Figure 35, the contributed reactive power will change accordingly. The droop curve describes the real converter response to voltage changes and is typically 2-4 percent.

The attribute VsConverter.qShare describe the sharing between parallel converters and is not a droop coefficient which makes it difficult to combine reactive contributions from other reactive sources, e.g. shunts, synchronous machines and static var compensators (SVC). Of the sources the SVC has droop control of the voltage.

Further clarifications of the CIM for this is needed, see Annex A point 24.

7.5 Attributes exchanged in CGM BP

The tables in this section lists the attributes from IEC TS 61970-600:2017 [1] that are exchanged in a HVDC IGM. No attributes from IEC 61970-600:2021 [7] are included.

Annex B list the changes in DC that has been made in IEC 61970-600:2021 [7] relative to IEC TS 61970-600:2017 [1]. IEC 61970-600:2021 DC mainly contain improved descriptive texts and additional restrictions. The restrictions may be implemented by CGM BP if determined to be needed. Implemented restrictions will be described in coming DC upgrade of QoCDCv3.2 [3].

Table 3 ACDCConverter

Attribute name	Exchanged	Profile	Comment
baseS	yes	EQ	
idleLoss	yes	EQ	
maxUdc	yes	EQ	
minUdc	yes	EQ	
numberOfValves	yes	EQ	This attribute is 6 for a 6-pulse and 12 for a 12-pulse ACDCConverter.
ratedUdc	yes	EQ	
resistiveLoss	yes	EQ	
switchingLoss	yes	EQ	
valveU0	no	EQ	

p	yes	SSH	
q	yes	SSH	This value shall be approximately p/2 for a current source converter.
targetPpcc	yes	SSH	This value shall agree with p at the rectifier and is used with the control modes CsPpccControlKind.activePower or VsPpccControlKind.pPcc control. Load sign convention is used.
targetUdc	yes	SSH	This value shall agree with udc at the inverter and is used with the control mode CsPpccControlKind.dcVoltage or VsPpccControlKind.udc.
poleLossP	yes	SV	
idc	yes	SV	
udc	yes	SV	
uc	no	SV	

1515

1516 **Table 4 ACDCConverterDCTerminal**

Attribute name	Exchanged	Profile	Comment
polarity	yes	EQ	

1517

1518 **Table 5 CsConverter**

Attribute name	Exchanged	Profile	Comment
maxAlpha	no	EQ	
maxGamma	no	EQ	
maxIdc	yes	EQ	
minAlpha	no	EQ	
minGamma	no	EQ	
minIdc	yes	EQ	
ratedIdc	yes	EQ	
operatingMode	yes	SSH	The rectifier delivers active power to the inverter. This is the case for both current and voltage source converters.
pPccControl	yes	SSH	The control mode CsPpccControlKind.activePower is used for the rectifier and

			CsPpccControlKind.dcVoltage for the inverter.
targetAlpha	no	SSH	
targetGamma	no	SSH	
targetIdc	no	SSH	
alpha	no	SV	
gamma	no	SV	

1519

1520 **Table 6 DCConverterUnit class**

Attribute name	Exchanged	Profile	Comment
operationMode	yes	EQ	

1521

1522 **Table 7 DCSeriesDevice class**

Attribute name	Exchanged	Profile	Comment
inductance	no	EQ	
resistance	yes	EQ	
ratedUdc	no	EQ	

1523

1524 **Table 8 DCShunt class**

Attribute name	Exchanged	Profile	Comment
capacitance	no	EQ	
resistance	yes	EQ	
ratedUdc	no	EQ	

1525

1526 **Table 9 VsConverter class**

Attribute name	Exchanged	Profile	Comment
maxModulationIndex	no	EQ	
maxValveCurrent	no	EQ	
droop	no	SSH	
droopCompensation	no	SSH	
pPccControl	yes	SSH	The control mode VsPpccControlKind.pPcc is used for the rectifier and VsPpccControlKind.udc for the inverter.

qPccControl	yes	SSH	Only the control mode VsQpccControlKind.voltagePcc is used.
qShare	yes	SSH	
targetQpcc	no	SSH	This value is used with the control mode VsQpccControlKind.reactivePcc. Load sign convention is used.
targetUpcc	yes	SSH	This value is used with the control mode VsQpccControlKind.voltagePcc.
delta	no	SV	
uc	no	SV	

1527

1528 **Table 10 Table 11 DCLineSegment class**

Attribute name	Exchanged	Profile	Comment
capacitance	no	EQ	
inductance	no	EQ	
resistance	yes	EQ	
length	no	EQ	

1529

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Annex A Proposed changes to CIM/CGMES

Based on the discussion in this document the following changes to the CIM standard are suggested

1. Add a new class that describe a HVDC Link.
2. Add a new class that describe a HVDC Pole. The class is contained by a HVDC Link and associates with the DCConverterUnits in the HVDC Pole. The boundary points, currently in cim:Line, move into the new HVDC Pole class.
3. To avoid the cases where substations need to be split in two where one is fictitious, may be avoided by moving the substations into the boundary. A single substation can the contain equipment from both the AC and DC IGMs.
4. Add a new class that allocate filters for compensating reactive power at current source converters and filter harmonics. The class need an association with the cim:RegulatingControl that have the shunts. The filter control function that allocates filters will use this class depending on the need for harmonics filtering. It use the cim:RegulatingCondEq.controlEnabled flags tell the voltage control which filters are available for voltage control.
5. If the filter control function is located as the DC side of the boundary it needs a path from the cim:ACDConverter to the cim:Regulating control doing the voltage control. This result in restrictions on the cim:ACDConverter.PccTerminal and cim:RegulatingControl.Terminal to be at cim:Teminals linked to the boundary nodes. This can be avoided if the filter control function is at the AC side.
6. The BPPL requirements section in IEC 61970-600-1 is suggested to be updated as discussed below
   a) BBPL6 states that “In the simplified exchange of an HVDC link the net interchange between the MAS is represented by EquivalentInjection classes referring to each common coupling node (CC)”. This is also true for detailed HVDC models which is suggested to be made explicit. The point of common coupling for a ACDCConverter it a Terminal which can be any Terminal. It is suggested to specify that it shall be the Terminal at the EquivalentInjection connected to the boundary node.
   b) BPPL8 states that a detailed HVDC model may be exchanged as a separate IGM or by included in the AC IGM. The detailed HVDC model shall always be exchanged as a separate detailed DC IGM that cannot be part of the AC IGM. A simplified HVDC model is described in the AC IGMs without the DC IGMs. A detailed HVDC link is described by also including the DC IGM in a CGM. This is done regardless if a HVDC link is internal or external. See also section 7.3.4.
   c) BPPL9 states that the DC IGM shall refer to the common coupling points in the boundary set. The point of common coupling is a cim:Terminal referenced by ACDCConverter. CGM BP recommends that this is a branch cim:Terminal connected to the boundary point, see also section 6.3.
   d) BPPL10 states that multiple HVDC Links may be included in a DC IGM. CGM BP recommends that each DC IGM can only contain one HVDC link but a HVDC link may contain multiple poles meaning that multiple DC cables and converters are included.
7. ACDCConverter.switchingLoss documentation says, “Switching losses, relative to the base apparent power 'baseS'.” but in the formula for computing the loss it is multiplied by ACDCConverter.idc. The text need check and correction.

8. The description of the VsConverter.droop attribute may be improved, the “D” in the formula need clarification. The VsConverter droop attributes relate to active power which is needed for DC networks. This must be clearly described.
9. Droop control of voltage is common for VsConverters but attributes supporting this is missing in CIM and needs to be added.
10. The description of the VsConverter.targetPhasePcc and VsConverter.targetPWMfactor attributes are left out from the CGM BP. The attributes need clarification and discussion of the relevance in power flow calculations.
11. An information model that describe the party responsible for a shared resource is needed, e.g. two TSOs that have a joint HVDC Link.
12. Figure 38 in IEC 61970-301 does not describe the desired boundary configuration for a HVDC Bipole and it is suggested the figure is replaced.
13. The formula for the pole loss is showed in the ACDCConverter.poleLossP attribute but it would be good to have it at a more prominent place, e.g. in the 301 text.
14. A formula that describe the curve in Figure 30 shall be added. The curve is a function of active power transfer at a given firing or extinction angle and is simple at a fix angle. The approximation Q=P/2 is a last resort that is always available.
15. Add that generation sign convention is used for ACDCConverter.maxP and ACDCConverter.minP
16. Add new classes that describe the bridge and the valves. The lack of a bridge class has resulted in confusion with the ACDCConverterUnit that instead has been used to represent the bridge. As the ACDCConverter unit represents the control of one side of a HVDC Pole rather than the bridges the description in the class need update.
17. The enumeration values activePower, dcVoltage and dcCurrent in CsPpccControlKind need to be investigated as they are probably related to different levels of control. The activePower relate to the whole HVDC Pole while dcVoltage and dcCurrent relate to each CsConverter in the HVDC Pole.
18. The enumeration values of VsPpccControlKind need also be investigated for the same reason. The VsQpccControlKind.pulseWidthModulation is probably a lower level control that shall not be mixed with the reactePower or voltage controls.
19. Text and diagrams in the HVDC section of IEC 61970-301:2020 are not aligned on parameter naming and the mathematical model describing the static power flow model need better explanation.
20. The control function that describe the dc Voltage at the CSC inverter as a function of transferred power need to be described in the standard. More parameters are probably needed in the UML for this. If the function is complex it may be described tabularly instead, this also requires a UML change. The function is described in the dynamics model and may be derived from it.
21. The control function that describe the firing/extinction angles and tap position as a function of a dc Voltage target is needed. More parameters are probably needed in the UML for this. If the function is complex it may be described tabularly instead, this also requires a UML change. The function is described in the dynamics model and may be derived from it.
22. A VsConverter link may have an operation mode where the link exhibits a variable series impedance to the AC system. This require that the series impedance r and x values be dynamically computed and are made available to power flow as attributes. The formulas for computing the r and x values need to be defined.

23. Has the linear loss term in the VSC loss equation the same value as the quadratic?
24. If more than one VSC controls the voltage at their PCCs the attribute VsConverter.qShare tells how the reactive injection shall be shared between the VSCs. This is not clear from the description. The attribute VsConverter.qShare is used with the control mode VsQpccControlKind.voltagePcc which could also be clarified. If multiple VSCs are connected to the AC substation it is assumed that they shall have the same reactive power control mode and the same target values but may have different VsConverter.qShare. This need also be described. Possibly could a RegulatingControl be used to provide common voltage targets would be a change in future CIM versions. In the case other reactive sources are used to control the same voltage a method to share the reactive production between the sources is needed, e.g. the SVC use droop control for this. As HVDC Poles, at least today, are operated at the same bus in the AC system, the bus split issue with RegulatingControl is not an issue for HVDC.
25. Variable impedance control mode is not supported in CIM and needs to be added in future versions, see also section 7.4.2.
26. The HVDC Bipole need to be described. A HVDC Bipole contain two HVDC Poles and has a control function that coordinates the two HVDC Poles. A solution is to describe it with a A HVDC Bipole class that will then be contained by the HVDC Link. See also items 1 and 2.

These changes are not intended for go-live of the CGM Building Process. It is suggested they are made in next CIM release, i.e. CIM18.

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Annex B HVDC in IEC 61970-600

IEC 61970-301:2020 Edition 7 and IEC 61970-600 has made changes to the HVDC information model from IEC 61970-301:2016 Edition 6. The changes are described in this annex.

To investigate the possible advantage of using the modelling improvement made in IEC 61970-600:2021 HVDC information model rather than using the IEC TS 61970-600:2017 HVDC information model. Most improvements in IEC 61970-600:2021 are descriptions that better aid the understanding of classes, attributes, and roles. Some improvements are additional restrictions on data which are important for implementations.

The DC information model from IEC 61970-600:2021 will not be used in CGM BP which makes this annex out of scope for CGM BP. But the annex is kept as a recording of work done based on the judgement that might still be useful as input to future improvements and for the purpose of completeness.

The DC changes in IEC 61970-301:2020 Edition 7 and IEC 61970-600 are listed in tables below and the column "Incl" stands for include in the CGM BP implementation. A "Yes" in the column means the change shall be included and added as a business process rule and restriction to the IEC TS 61970-600 based information model used for CGM BP. A "No" in the "Incl" column means that the attribute is not used in power flow and need not be exchanged for the current scope of CGM build process

The tables below describe the information that shall be used in implementations, as a result of detailed analysis, for the purpose of steady state analysis and power flow.

The "Prfl" column indicates the profile.

Table 12 ACDCConverter class changes

Attribute/Role	Type of change	Change	Prfl	Incl
baseS	Description extended	The attribute shall be a positive value.	EQ	Yes
idleLoss	Description extended	The attribute shall be a positive value.	EQ	Yes
maxUdc	Description extended	The attribute shall be a positive value.	EQ	Yes
minUdc	Description extended	The minimum voltage on the DC side at which the converter should operate. It is converters configuration data used in power flow. The attribute shall be a positive value.	EQ	Yes
poleLossP	Description extended	The attribute shall be a positive value.	SV	Yes
ratedUdc	Description extended	The attribute shall be a positive value.	EQ	Yes

		For instance, a bipolar HVDC link with value 200 kV has a 400kV difference between the dc lines		
resistiveLoss	Description extended	The attribute shall be a positive value.	EQ	Yes
switchingLoss	Description extended	The attribute shall be a positive value.	EQ	Yes
targetPpcc	Description extended	Load sign convention is used, i.e. positive sign means flow out from a node.	SSH	Yes
targetPpcc	Cardinality change	Optional	SSH	Yes
targetUdc	Description extended	The attribute shall be a positive value.	SSH	Yes
targetUdc	Cardinality change	Optional	SSH	Yes
uc	Description extended	Line-to-line converter voltage the voltage at the AC side of the valve. It is converters state variable, result from power flow. The attribute shall be a positive value. The attribute shall be a positive value.	SV	No
udc	Description extended	The attribute shall be a positive value.	SV	Yes
valveU0	Description extended	also called Uvalve.	EQ	No
maxP	Added attribute	Maximum active power limit. The value is overwritten by values of VsCapabilityCurve, if present. An active power order shall be below this limit.	EQ	No
minP	Added attribute	Minimum active power limit. The value is overwritten by values of VsCapabilityCurve, if present. An active power order shall be above this limit.	EQ	No
ACDCConverterDCTerminal	Add role descriptions	A DC converter have DC converter terminals. A converter has two DC converter terminals.	EQ	Yes

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PccTerminal	Delete text	The power flow measurement must be the sum of all flows into the transformer.	EQ	Yes

1659 **Table 13 ACDCConverterDCTerminal class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
polarity	Description extended	Depending on the converter configuration the value shall be set as follows: - For a monopole with two converter terminals use DCPolarityKind 'positive' and 'negative'. - For a bipole or symmetric monopole with three converter terminals use DCPolarityKind 'positive', 'middle' and 'negative '.	EQ	Yes
polarity	Cardinality change	Required	EQ	Yes

1660 **Table 14 CsConverter class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
-	Class description extended	The firing angle controls the dc voltage at the converter, both for rectifier and inverter. The difference between the dc voltages of the rectifier and inverter determines the dc current. The extinction angle is used to limit the dc voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source dc line. Higher level controls are built on top, e.g. dc voltage, dc current and active power. From a steady state perspective it is sufficient to specify the wanted active power transfer	NA	Yes

		(ACDCConverter.targetPpcc) and the control functions will set the dc voltage, dc current, firing angle, transformer tap position and number of connected filters to meet this. Therefore, attributes targetAlpha and targetGamma are not applicable in this case. The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses is a function of the dc voltage and dc current. The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10-18 degrees. The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17-20 degrees.		
alpha	Description extended	Firing angle that determines the dc voltage at the converter dc terminal. The attribute shall be a positive value.	SV	No
gamma	Description extended	It is used to limit the dc voltage at the inverter if needed. Typical value between 17 degrees and 20 degrees for an inverter. The attribute shall be a positive value.	SV	No

maxAlpha	Description extended	The attribute shall be a positive value.	EQ	No
maxGamma	Description extended	The attribute shall be a positive value.	EQ	No
minIdc	Description extended	The attribute shall be a positive value.	EQ	Yes
pPccControl	Description added	Kind of active power control.	SSH	Yes
ratedIdc	Description extended	The attribute shall be a positive value.	EQ	Yes
targetAlpha	Description extended	It is only applicable for rectifier if continuous tap changer control is used. Allowed values are within the range minAlpha<=targetAlpha<=maxAlpha. The attribute shall be a positive value.	SSH	No
targetGamma	Description extended	It is only applicable for inverter if continuous tap changer control is used. Allowed values are within the range minGamma<=targetGamma<=maxGamma. The attribute shall be a positive value.”	SSH	No
targetIdc	Description extended	The attribute shall be a positive value.	SSH	No
CSCDynamics	Association added for DY profile.		DY	No

1661 **Table 15 CsOperatingModeKind class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
inverter	Description extended	which is the power receiving end.	SSH	Yes
rectifier	Description extended	which is the power sending end.	SSH	Yes

1662 **Table 16 CsPpccControlKind class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
activePower	Description changed	Control is active power control at AC side, at point of common coupling. Target is provided by ACDCConverter.targetPpcc.	SSH	Yes
dcVoltage	Description changed	Control is DC voltage with target value provided by ACDCConverter.targetUdc.	SSH	Yes
dcCurrent	Description changed	Control is DC current with target value provided by CsConverter.targetIdc.	SSH	No

1663 **Table 17 DCBaseTerminal class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
DCNode	Description changed	The DC connectivity node to which this DC base terminal connects with zero impedance.	EQ	Yes

1664 **Table 18 DCConductingEquipment class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
ratedUdc	New attribute	Rated converter DC voltage, also called UdN. The attribute shall be a positive value. It is converters configuration data used in power flow. For instance, a bipolar HVDC link with value 200 kV has a 400kV difference between the dc lines.	EQ	No
DcTerminal	Description added	A DC conducting equipment has DC terminals.	EQ	Yes

1665 **Table 19 DCConverterUnit class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
operationMode	Description added	The operating mode of an HVDC Bipole (bipolar, monopolar metallic return, etc).	EQ	Yes
Substation	Description added	The containing substation of the DC converter unit.	EQ	Yes

1666 **Table 20 DCEquipmentContainer class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
DCNode	Description added	The DC nodes contained in the DC equipment container.	TP	Yes
DCTopologicalNode	Description added	The topological nodes which belong to this connectivity node container.	TP	Yes

1667 **Table 21 DCLine class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
Region	Description added	The SubGeographicalRegion containing the DC line.	EQ	Yes

1668 **Table 22 DCNode class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
DCTopologicalNode	Description added	The DC topological node to which this DC connectivity node is assigned. May depend on the current state of switches in the network.	TP	Yes
DCTerminals	Description added	DC base terminals interconnected with zero impedance at a this DC connectivity node.	EQ	Yes

1669 **Table 23 DCPolarityKind class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
positive	Description extended	The converter terminal is intended to operate at a positive voltage relative the midpoint or negative terminal.	EQ	Yes
middle	Description extended	The converter terminal is the midpoint in a bipolar or symmetric monopole configuration. The midpoint can be grounded and/or have a metallic return.	EQ	Yes
negative	Description extended	The converter terminal is intended to operate at a negative	EQ	Yes

		voltage relative the midpoint or positive terminal.		

1670 **Table 24 DCSeriesDevice class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
ratedUdc	Deleted	inherited from: DCConductingEquipment	EQ	No

1671 **Table 25 DCShunt class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
ratedUdc	Deleted	inherited from: DCConductingEquipment	EQ	No

1672 **Table 26 DCTopologicalIsland class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
DCTopologicalNode	Description added	The DC topological nodes in a DC topological island.	TP	Yes

1673 **Table 27 VsConverter class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
delta	Description extended	The attribute shall be a positive value or zero.	SV	No
droop	Description extended Unit changed	The attribute shall be a positive value. modified the unit to PU	SSH	No
droop	Cardinality change	Optional	SSH	No
droopCompensation	Description extended	clarified that it is resistance The attribute shall be a positive value.	SSH	No
droopCompensation	Cardinality change	Optional	SSH	No
maxValveCurrent	Attribute deleted		NA	No
qPccControl	Description added	Kind of reactive power control.	SSH	Yes
qShare	Description extended	The attribute shall be a positive value or zero.	SSH	Yes

qShare	Cardinality change	Optional	SSH	Yes
targetQpcc	Description extended	Load sign convention is used, i.e. positive sign means flow out from a node.	SSH	Yes
targetQpcc	Cardinality change	Optional	SSH	Yes
targetUpcc	Description extended	The attribute shall be a positive value.	SSH	Yes
targetUpcc	Cardinality change	Optional	SSH	Yes
uv	Renamed from uf Description changed	Line-to-line voltage on the valve side of the converter transformer. It is convert's state variable, result from power flow. The attribute shall be a positive value.	SV	No
targetPowerfactorPcc	New attribute	Power factor target at the AC side, at point of common coupling. The attribute shall be a positive value.	SSH	No
targetPhasePcc	New attribute	Phase target at AC side, at point of common coupling. The attribute shall be a positive value.	SSH	No
targetPWMfactor	New attribute	Magnitude of pulse-modulation factor. The attribute shall be a positive value.	SSH	No
VSCDynamics	New role in DY profile		DY	No
maxModulationIndex	No change		SSH	No
maxValvecurrent	No change		EQ	No

1674 **Table 28 VsPpccControlKind class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
pPcc	Description extended	The target value is provided by ACDCConverter.targetPpcc.	SSH	Yes
udc	Description changed	with target value provided by ACDCConverter.targetUdc.	SSH	Yes

pPccAndUdcDroop	Description extended	Target values are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.	SSH	No
pPccAndUdcDroop WithCompensation	Description extended	Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc, VsConverter.droop and VsConverter.droopCompensation.	SSH	No
pPccAndUdcDroop Pilot	Description extended	The mode is used for Multi Terminal High Voltage DC (MTDC) systems where multiple HVDC Substations are connected to the HVDC transmission lines. The pilot voltage is then used to coordinate the control the DC voltage across the HVDC substations. Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.	SSH	No
phasePcc	New attribute	Control is phase at point of common coupling. Target is provided by VsConverter.targetPhasePcc.	SSH	No

1675 **Table 29 VsQpccControlKind class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
-	Class description added	Types applicable to the control of real power and/or DC voltage by voltage source converter.	SSH	Yes
reactivePcc	Description extended	Control is reactive power at point of common coupling. Target is provided by VsConverter.targetQpcc.	SSH	Yes
voltagePcc	Description extended	Control is voltage at point of common coupling. Target is provided by VsConverter.targetUpcc.	SSH	Yes
powerFactorPcc	Description extended	Control is power factor at point of common coupling. Target is	SSH	Yes

		provided by VsConverter.targetPowerFactorPcc.		
pulseWidthModulation	New attribute	No explicit control. Pulse-modulation factor is directly set in magnitude (VsConverter.targetPWMfactor) and phase (VsConverter.targetPhasePcc).	SSH	No

1676 **Table 30 DCLineSegment class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
capacitance	Cardinality change	Required	EQ	Yes
resistance	Description extended	The attribute shall be a positive value.	EQ	Yes

1677 **Table 31 PerLengthDCLineParameter class changes**

Attribute/Role	Type of change	Change	Prfl	Incl
-	Class deleted		EQ	Yes

1678 **Table 32 BoundaryPoint class changes (not in DC)**

Class/Attribute/Role	Type of change	Change	Prfl	Incl
BoundaryPoint		New class		No
isDirectCurrent	Used in DC		EQ& TP	No
isExcludedFromAreaInterchange	Used in DC		EQ& TP	No

1679 **Table 33 Class rules**

Class	Attribute/Role	Rule	Incl
DCGround	-	At least one DCGround shall be defined for each dc circuit	Yes
Equipment	EquipmentContainer	ProtectedSwitch (Breaker, DisconnectingCircuitBreaker, LoadBreakSwitch) the association shall point to EquipmentContainer of type Bay, Line or DCConverterUnit. SeriesCompensator the association shall point to EquipmentContainer of type VoltageLevel when in substation, DCConverterUnit or Line when outside substation.	

		PowerTransfomer the association shall point to EquipmentContainer of type Substation or DCConverterUnit. Disconnector the association shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation. GroundDisconnector the association shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation. Fuse the association shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation. Jumper the association shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation. Cut the association shall point to EquipmentContainer of type Bay, VoltageLevel or DCConverterUnit or Line when outside substation. DCSwitch (DCDisconnector, DCBreaker) the association shall point to EquipmentContainer of type DCConverterUnit. DCGround the association shall point to EquipmentContainer of type DCConverterUnit. DCBusbar the association shall point to EquipmentContainer of type DCConverterUnit. DCChopper the association shall point to EquipmentContainer of type DCConverterUnit. DCShunt the association shall point to EquipmentContainer of type DCConverterUnit.	

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		DCSeriesDevice the association shall point to EquipmentContainer of type DCConverterUnit. DCLineSegment the association shall point to EquipmentContainer of type DCLine. In the case of modelling back to back configuration the association shall point to EquipmentContainer of type Substation. ACDCConverter (CsConverter, VsConverter) the association shall point to DCEquipmentContainer of type DCConverterUnit. In this case the association DCConverterUnit.Substation is required.	
EquivalentInjection		If EquivalentInjection.regulationCapability in EQ is true, then EquivalentInjection.regulationStatus and EquivalentInjection.regulationTarget are required in SSH. If EquivalentInjection.regulationCapability in EQ is false, then EquivalentInjection.regulationStatus and EquivalentInjection.regulationTarget are not exchanged in SSH. If EquivalentInjection connects to a BoundaryPoint with flag isDirectCurrent=false (meaning this is not HVDC), the EquivalentInjection.regulationCapability in EQ shall be set to false and there shall not be a ReactiveCapabilityCurve associated.	
ACDCConverter		The target values and related attributes for ACDCConverter and its subclasses are optional in SSH. However, depending on the control mode of the converter some of the attributes shall be considered as required. The description of the control modes in the enumerations	

		CsPpccControlKind, VsPpccControlKind and VsQpccControlKind provide information on necessary attributes which are then considered required attributes for each control mode in SSH.	

As an advice, when eventually moving from IEC TS 61970-600:2016 [1] to IEC 61970-600:2021 [7] the following changes need attention:

- The attribute ratedUdc that is moved to the base class DCConductingEquipment that means a name change of this attribute in the EQ CIMXML file.

Annex C Scheduling internal HVDC Links

For internal HVDC, not controlling 1-2-1 the flow on an AC border, there is no obligation to exchange a schedule within the market data. In this case it is the TSO with the HVDC Link that transfers representative schedules to the HVDC injections in SSH.

When internal HVDC models are provided as a detailed model, this shall go into separate DC IGM with as consequence:

- Internal boundary nodes are to be used at the border of this DC IGM. A CR will be issued to NMD to allow the use of internal boundary nodes.
- EIC-T codes needs to be requested for the internal HVDC link in order to be able to make the link between the CIM:DCConverterUnit and CIM:line containing the HVDC boundary injections.

In the case the HVDC Link is close to the border a single or a group of tie lines may carry the power transferred by the HVDC Link. The effect of the HVDC Link is then that it forces the transfer over the tie lines. It is then important that the values from the HVDC schedule are properly transferred to the HVDC boundary injections (the EI.p attributes in SSH). If the HVDC boundary injections do not match the schedule values power flow may not be able to solve. For this reason, internal HVDC Link schedules must be provided for HVDC Links close to AC border.

Using HVDC schedules for close to AC border HVDC, TSO's can express that the flow on the AC border is controlled by the HVDC. This has not only an advantage during power flow, but also an advantage to express within CGMA computation that the flow on the nearby AC border is controlled. A CR will be issue to CGMA to enable this.

Internal HVDC schedule shall be provided rather than relying on cross-border schedule on the nearby AC border for following reasons:

- In case the small network from AC IGM 1 is meshed, it is impossible to reach the cross-border targets on AC boundary nodes.
- In case of injection (e.g. wind generation units) between HVDC and controlled AC border, a power flow would be necessary to know how much to inject on the HVDC.

This means concretely that within RMD/RIMD a VSA needs to be used always, either for the in or for the out Domain. The connectingLine_RegisteredResource shall refer to the EIC_T of HVDC link. Hence EIC_T code and EIC-Y code of the VSA are mandatory for close to border internal HVDC links

In case TieFlow objects are provided on the associated AC terminals connected to the AC boundary nodes, the flow on the AC boundary points will be assessed and considered inside AC NP.

To be able to define and assess the direction of the flow, additional information is needed to link the market data for each landing side per pole to the grid model. This is necessary as in/out domain are the same on both landing side per pole in market data. Hence a CR will be foreseen to include this additional information in the RIMD which is then to be provided for close to the border HVDC.

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